

Video Info

File name: P-CDL-I-M5-1-Trust and Security with Google Cloud

Presenter:



05

Trust and Security with Google Cloud

Module 5

Trust and Security with Google Cloud



Lessons

- 01 Trust and security in the cloud
- 02 Google's trusted infrastructure
- 03 Google Cloud's trust principles and compliance

Module 5

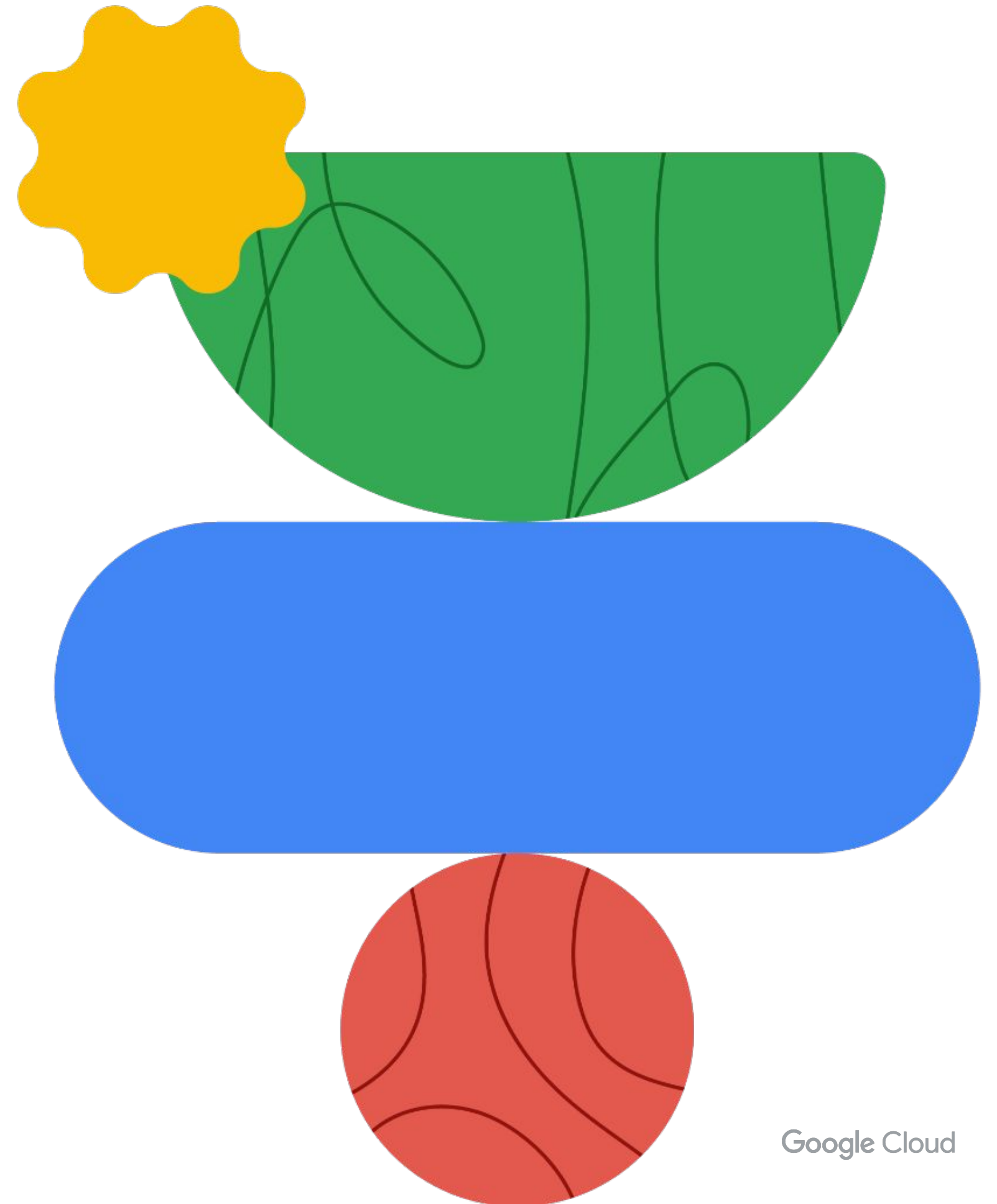
Trust and Security with Google Cloud



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Key security terms and concepts



Concepts related to reducing the risk of unauthorized access to sensitive data

Privileged access

Grants specific users access to a broader set of resources than ordinary users.

Least privilege

Advocates granting users only the access they need to perform their job responsibilities.

Zero-trust architecture

Assumes that no user or device can be trusted by default.

Concepts related to how an organization can protect itself from cyber threats

Security by default

Emphasizes integrating security measures into systems and applications from the initial stages of development.

Security posture

The overall security status of a cloud environment.

Cyber resilience

An organization's ability to withstand and recover quickly from cyber attacks.

Concepts related to protecting cloud resources from unauthorized access

Firewall

A network device that regulates traffic based on predefined security rules.

Encryption

The process of converting data into an unreadable format by using an encryption algorithm.

Decryption

Uses an encryption key to restore encrypted data back to its original form.

Quiz

Question

Which security principle advocates granting users only the access they need to perform their job responsibilities?

- A. Security by default
- B. Least privilege
- C. Zero-trust architecture
- D. Privileged access

Quiz

Answer

Which security principle advocates granting users only the access they need to perform their job responsibilities?

- A. Security by default
- B. Least privilege
- C. Zero-trust architecture
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Quiz

Question

Which definition best describes a firewall?

- A. A security model that assumes no user or device can be trusted by default
- B. A set of security measures designed to protect a computer system or network from cyber attacks
- C. A software program that encrypts data to make it unreadable to unauthorized users
- D. A network security device that monitors and controls incoming and outgoing network traffic based on predefined security rules

Quiz

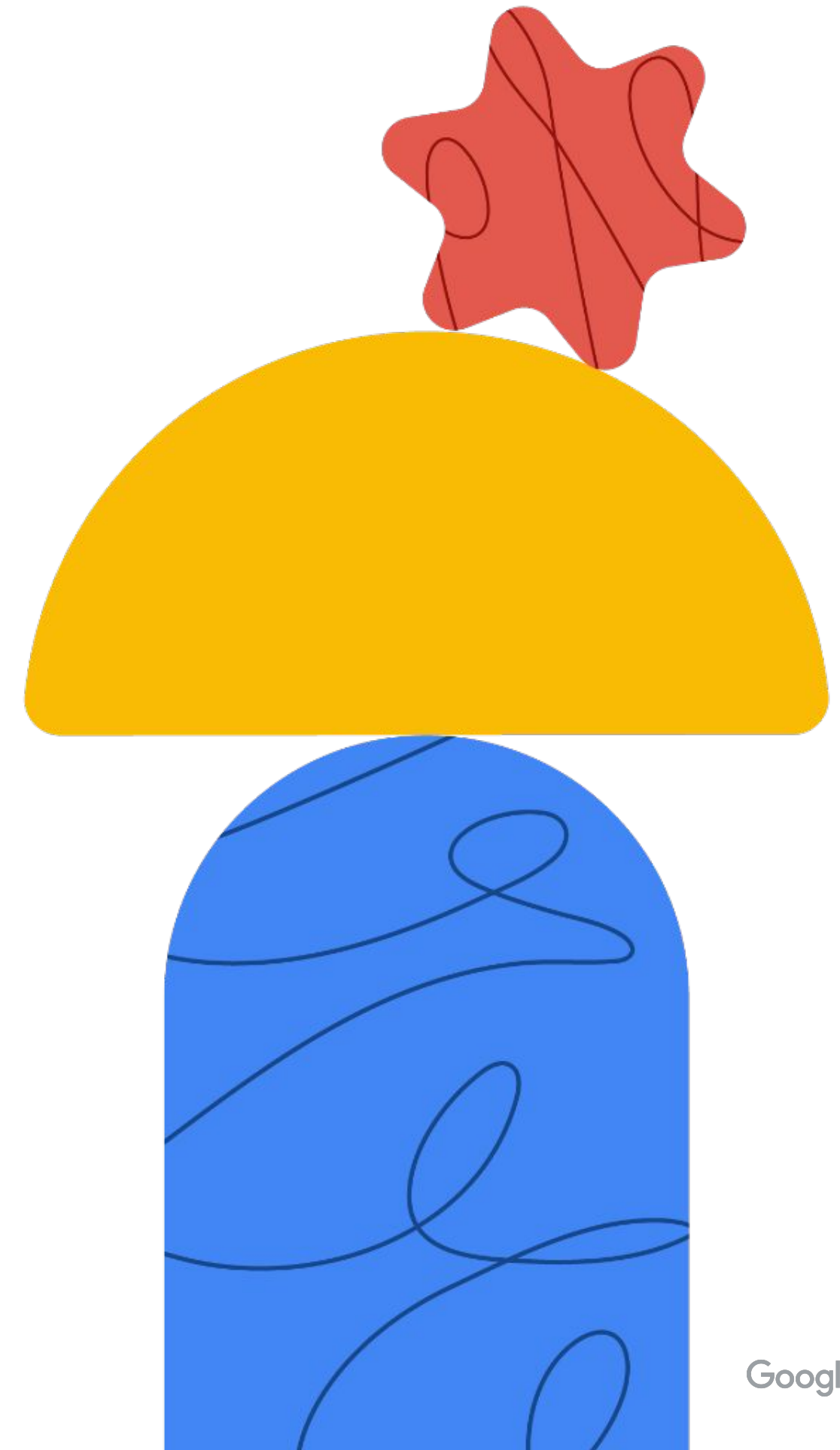
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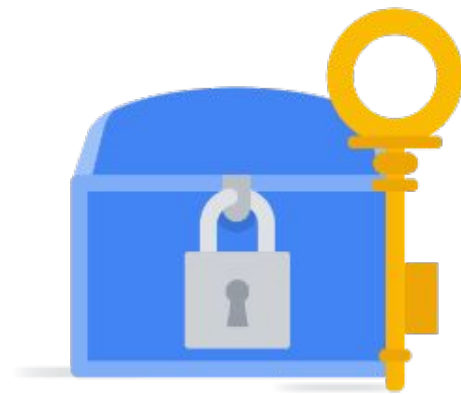
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Cloud security components



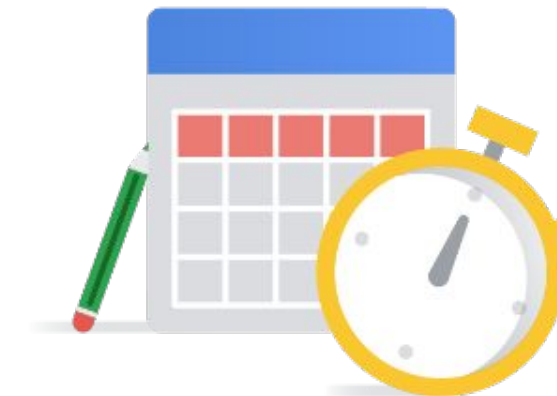
Three essential aspects of security



Confidentiality

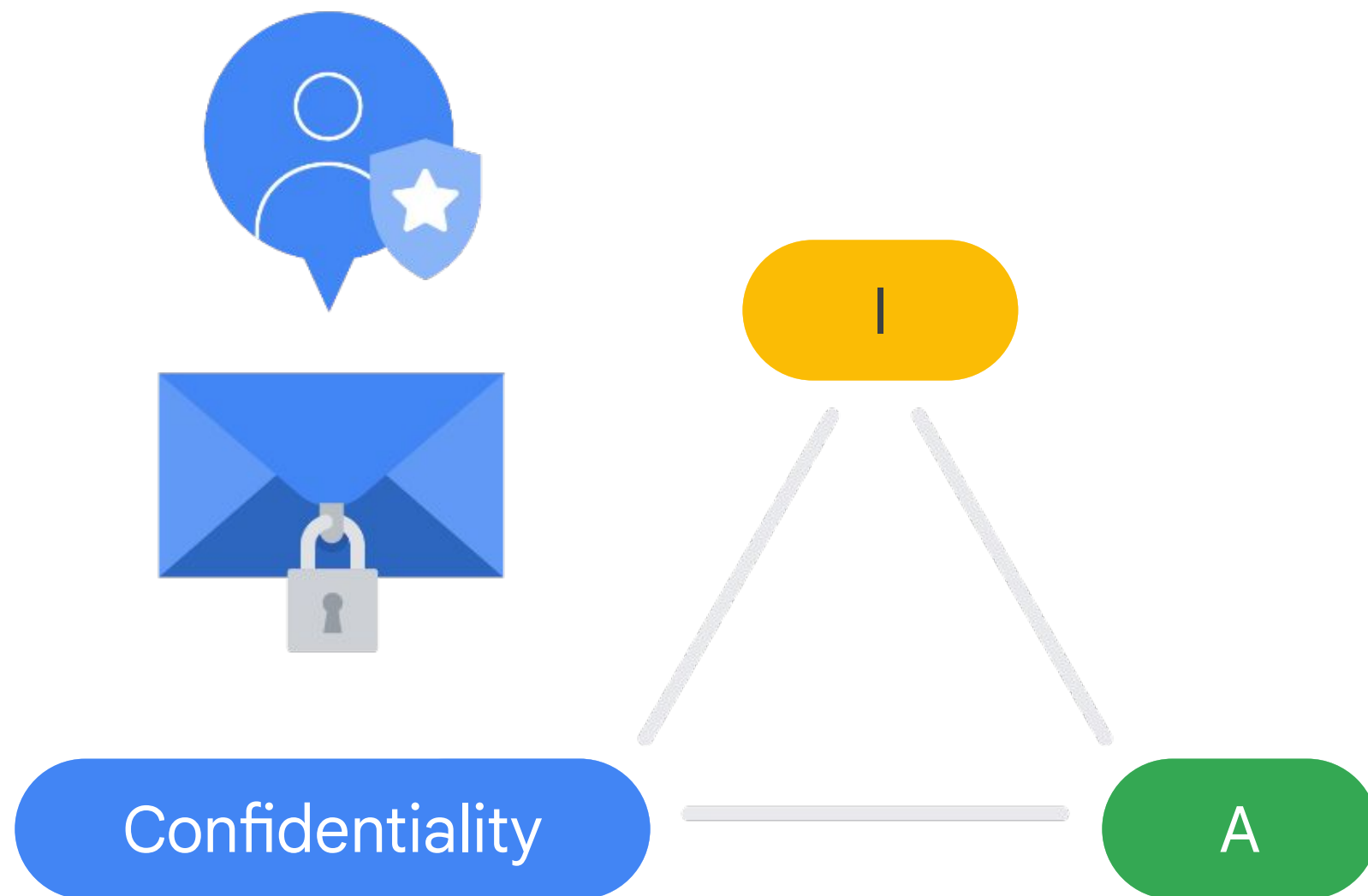


Integrity



Availability

Confidentiality



Confidentiality is about keeping important information safe and secret.

It ensures that only authorized people can access sensitive data, no matter where it's stored or sent.

Confidentiality + encryption



Encryption

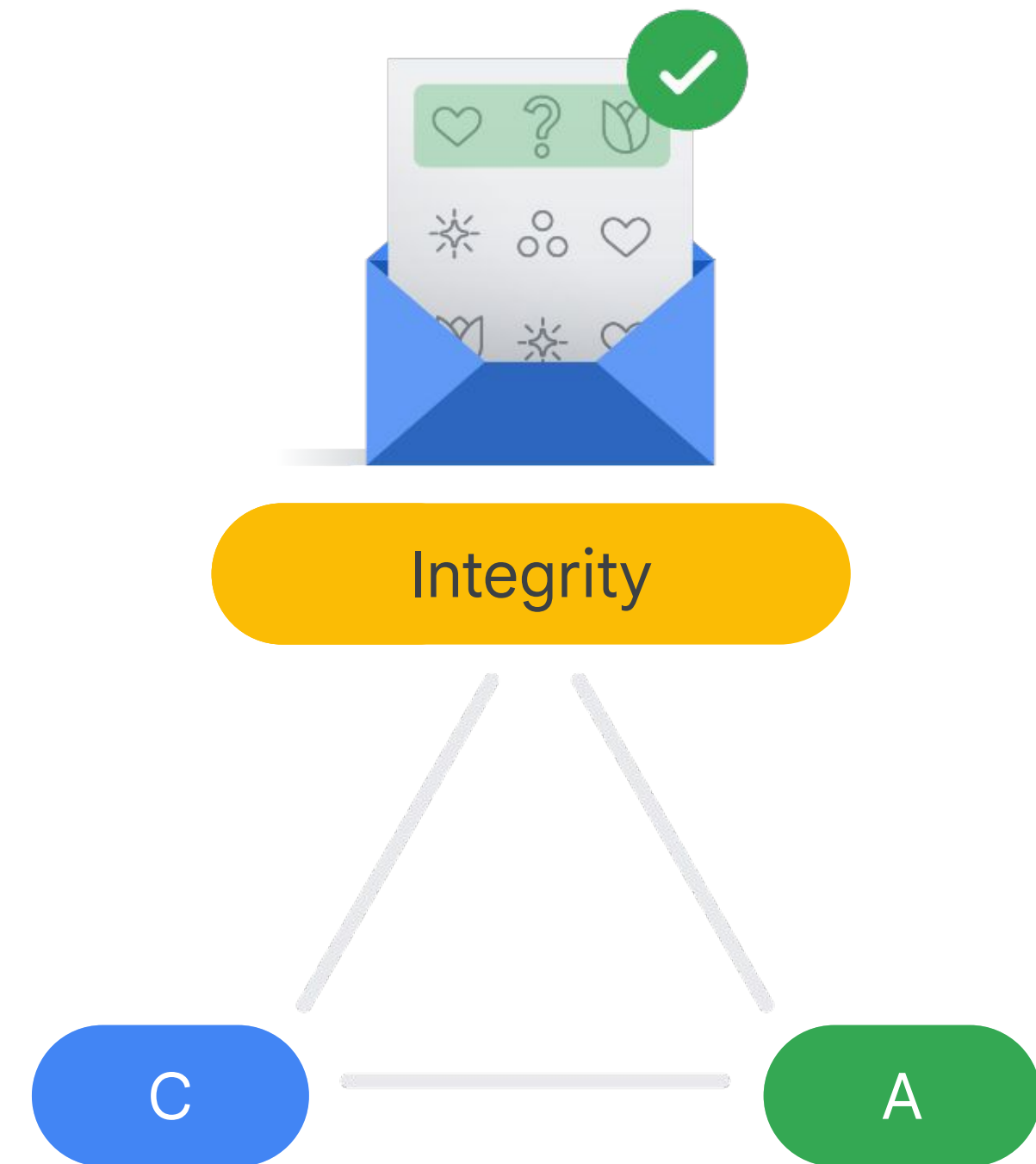


Encryption techniques and encryption keys help ensure that only authorized individuals can access and decrypt sensitive data.

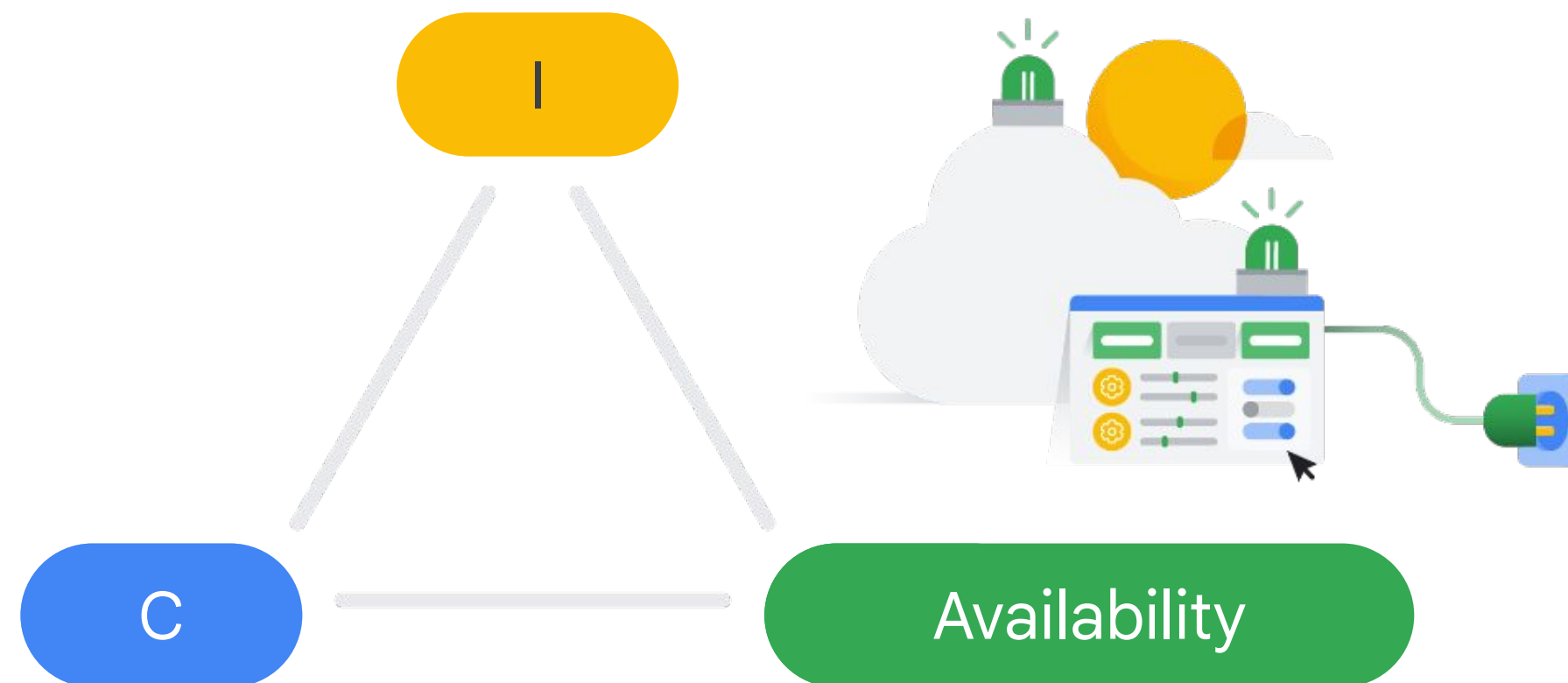
Integrity

Integrity means keeping data accurate and trustworthy.

It ensures that information doesn't get changed or corrupted, no matter where it's stored or how it's moved around.



Availability



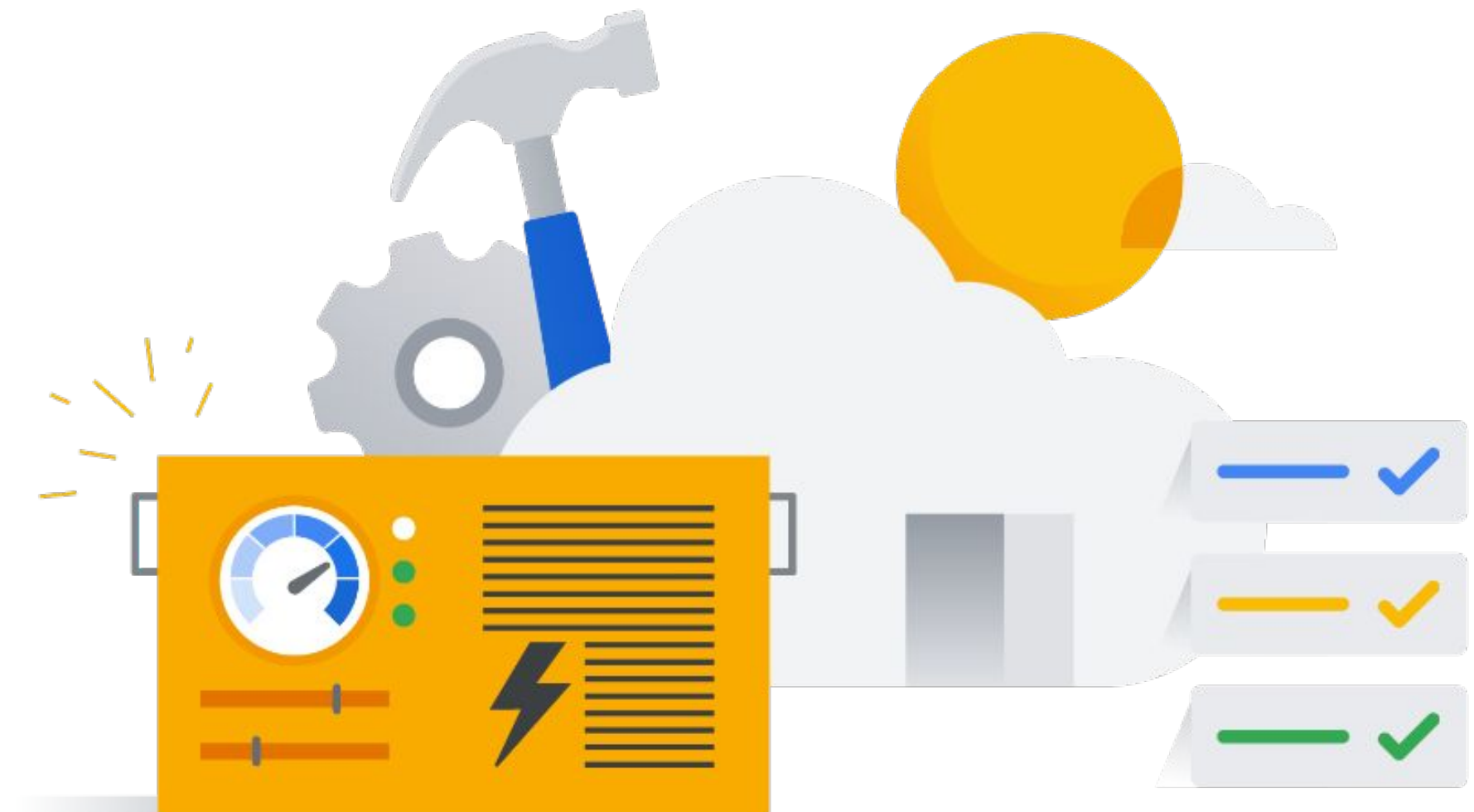
Availability is making sure that cloud systems and services are always accessible and ready for use by the right people when needed.

It's like having a reliable electricity supply that never goes out.

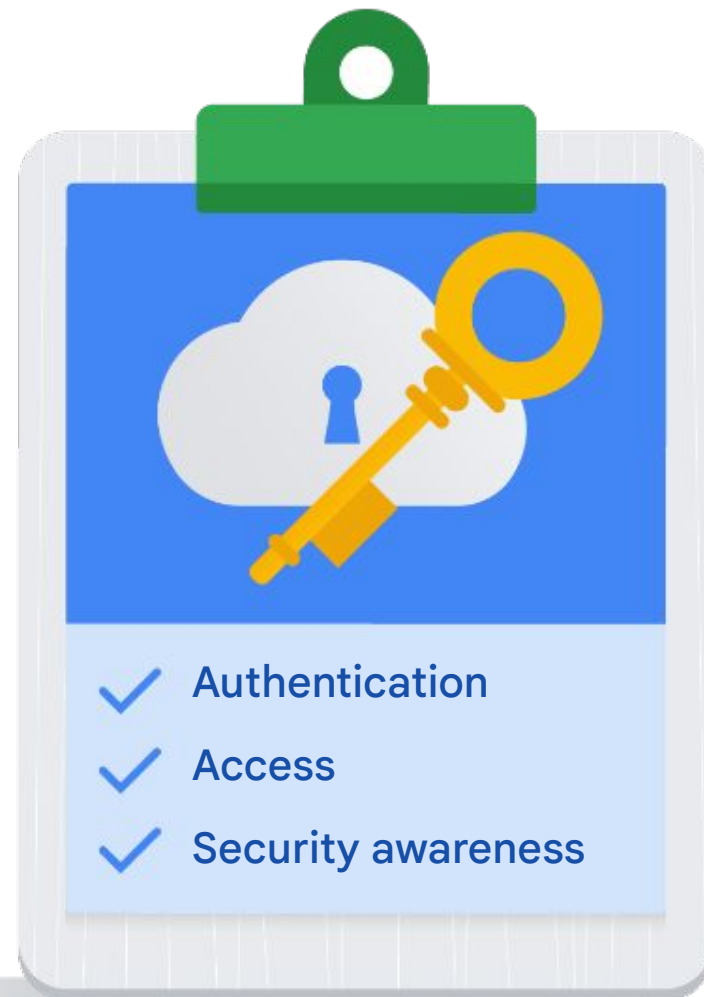
Availability measures can ensure that systems and applications in the cloud remain accessible

Cloud environments must be designed with:

- ✓ Redundancy
- ✓ Failover mechanisms
- ✓ Disaster recovery plans



Control

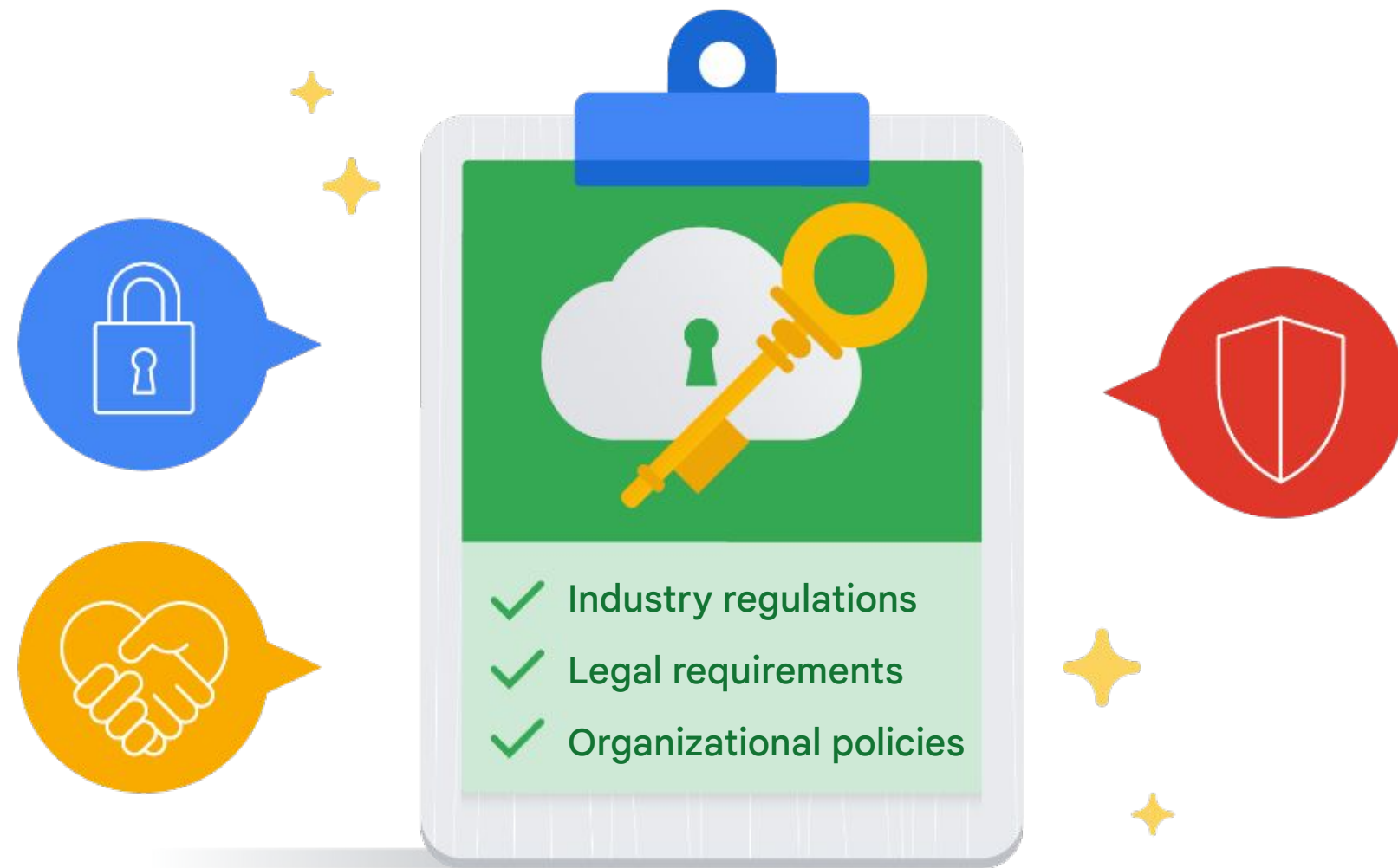


Control

The measures and processes implemented to manage and mitigate security risks.

Protect against unauthorized access, misuse, and potential threats.

Compliance



Compliance relates to adhering to industry regulations, legal requirements, and organizational policies.

It involves ensuring that security practices and measures align with established standards and guidelines.

Cloud providers often offer compliance frameworks and certifications.

Activity

Identify the cloud security concept that matches the description.

Establishing policies, procedures, and technical safeguards to protect against unauthorized access, misuse, and potential threats.

1

Making sure that cloud systems and services are always accessible and ready for use by the right people when needed.

2

Ensuring that security practices and measures align with established standards and guidelines.

3

Ensuring that information doesn't get changed or corrupted, no matter where it's stored or how it's moved around.

4

Ensuring that only authorized people can access sensitive data, no matter where it's stored or sent.

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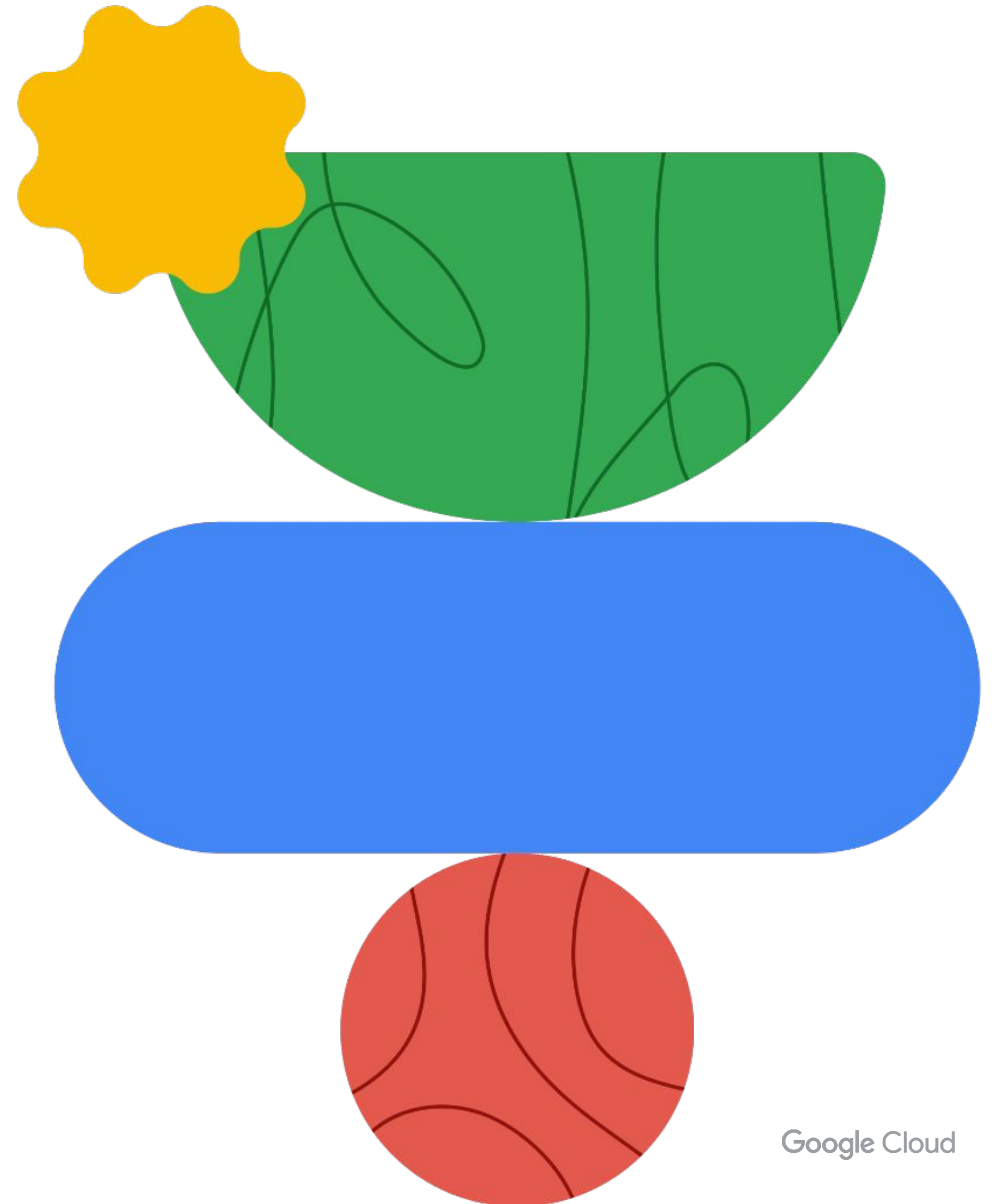
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Integrity

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Confidentiality

Cybersecurity threats



Social engineering

Anyone within your organization can be tricked into inadvertently downloading malicious attachments, divulging passwords, or compromising sensitive data.

Phishing attacks



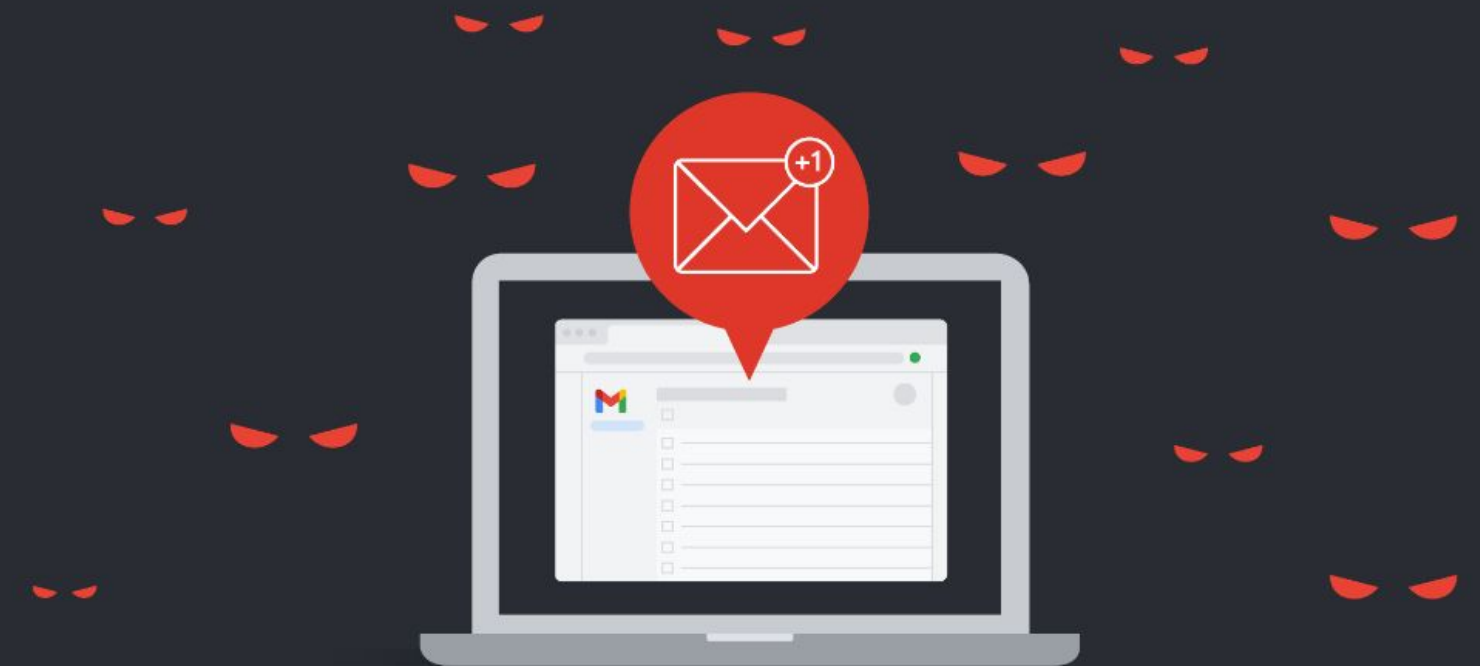
Collect personal details about you.



Collect personal details about your employees.



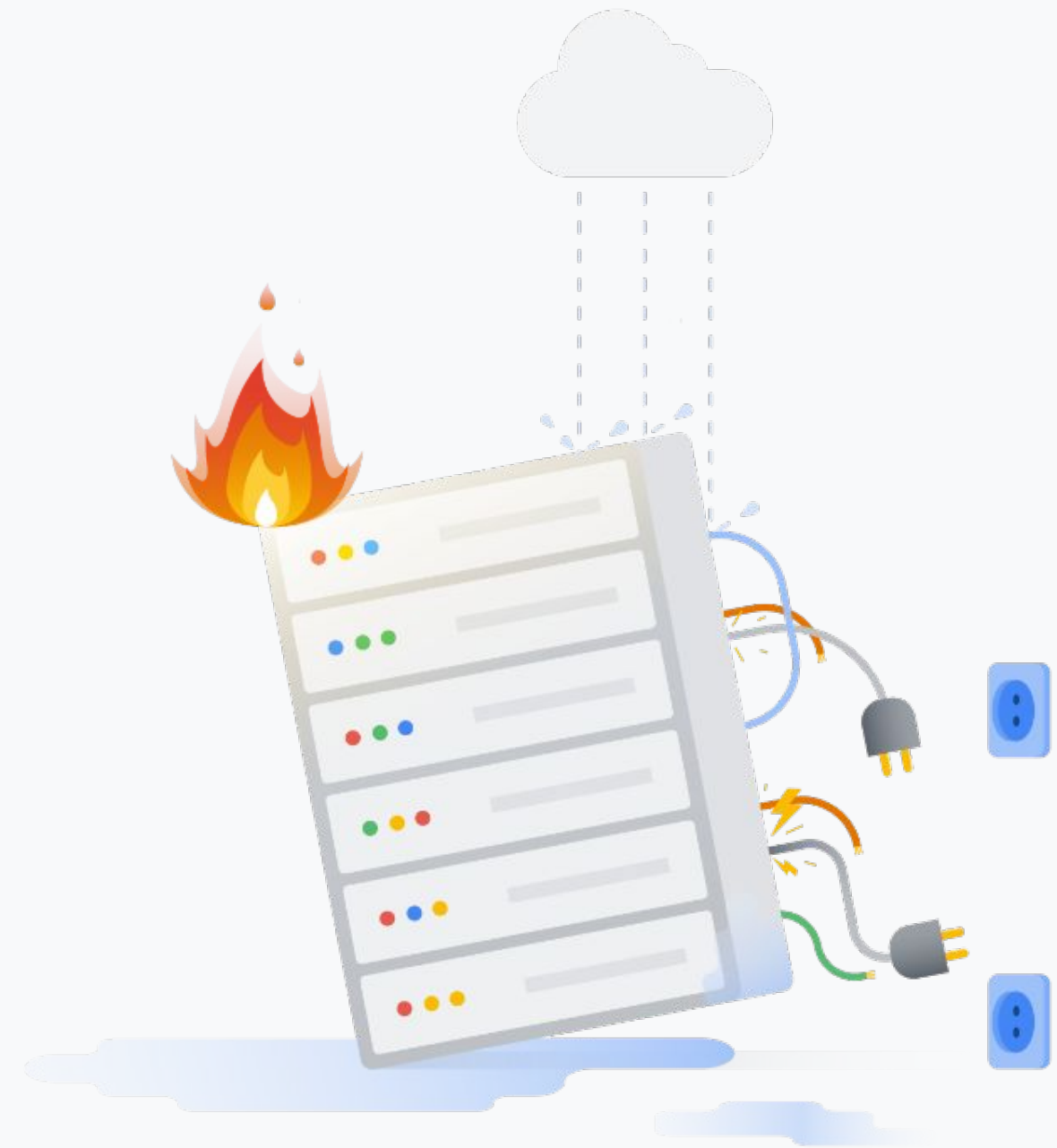
Collect personal details about your students.



Physical damage

Organizations are responsible for safeguarding data even in the face of physical adversity.

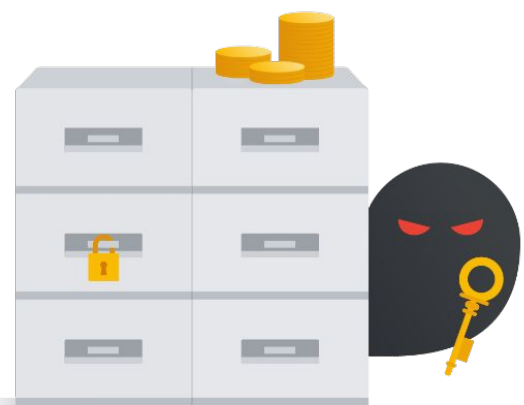
- ! Damage to hardware components
- ! Power disruptions
- ! Natural disasters: Floods, fires, and earthquakes



Malware, viruses, and ransomware

This malicious software aims to disrupt operations, inflict damage, or gain unauthorized access to computer systems.

Ransomware



Crucial files are held hostage until a considerable ransom is paid.

It's like witnessing the digital equivalent of an extortion scheme.



Vulnerable third-party systems

Without adequate security measures, third-party systems can transform into potential threats, leaving data security vulnerable.

Common functions for third-party systems:

- ✓ Finance
- ✓ Inventory management
- ✓ Account operations

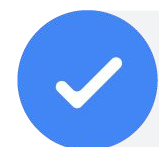


Configuration mishaps

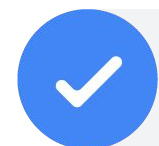
Misconfiguration occurs when errors arise during the setup or configuration of resources, which inadvertently exposes sensitive data and systems to unauthorized access.



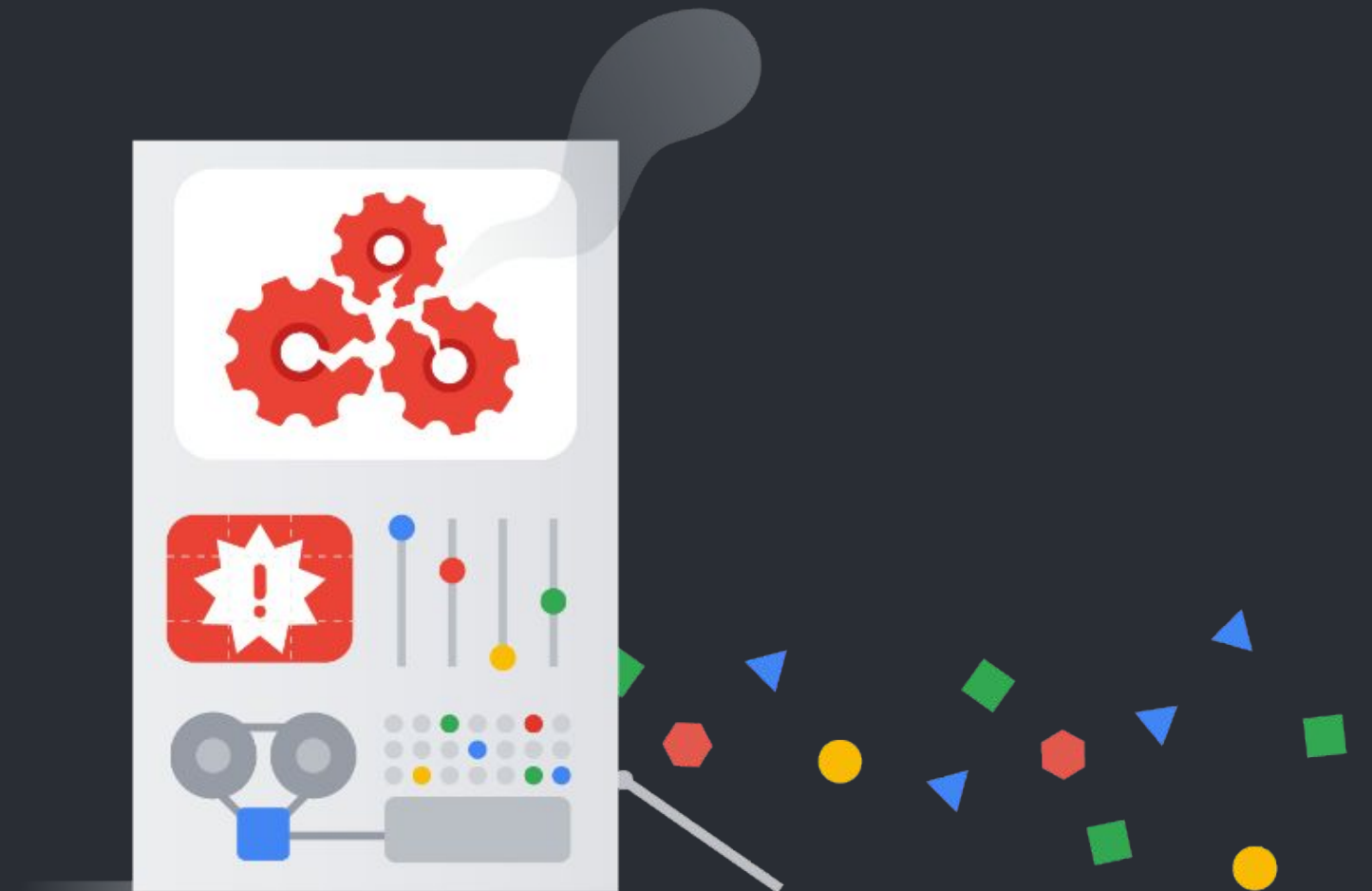
Surveys identify misconfiguration as the most prominent threat to cloud security.



Adopt principles of least privilege



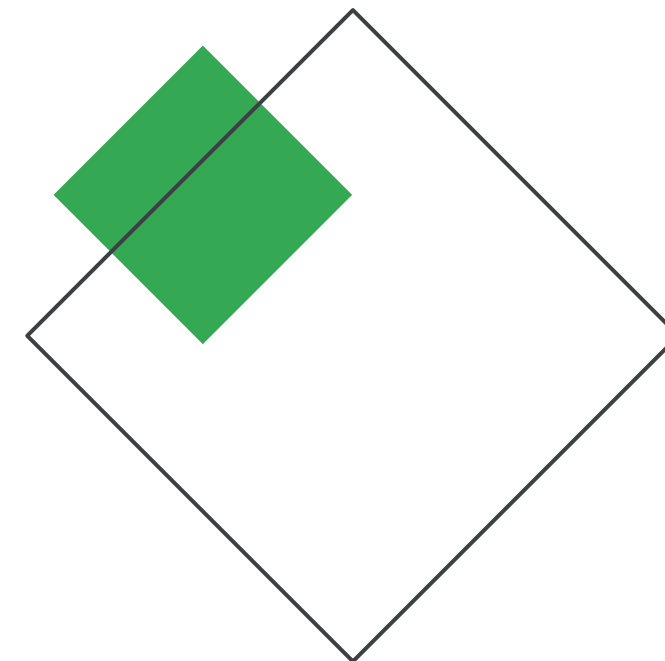
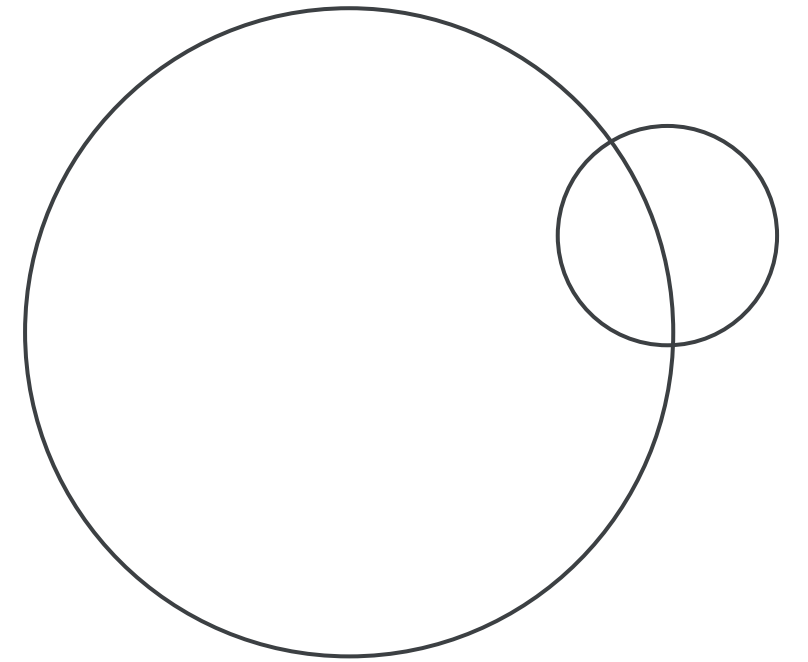
Adopt principles of privileged access



Example 1

An employee finds an old USB memory stick in his home and connects it to his work computer. Soon after, the device begins acting erratically, with many of his files being corrupted or deleted.

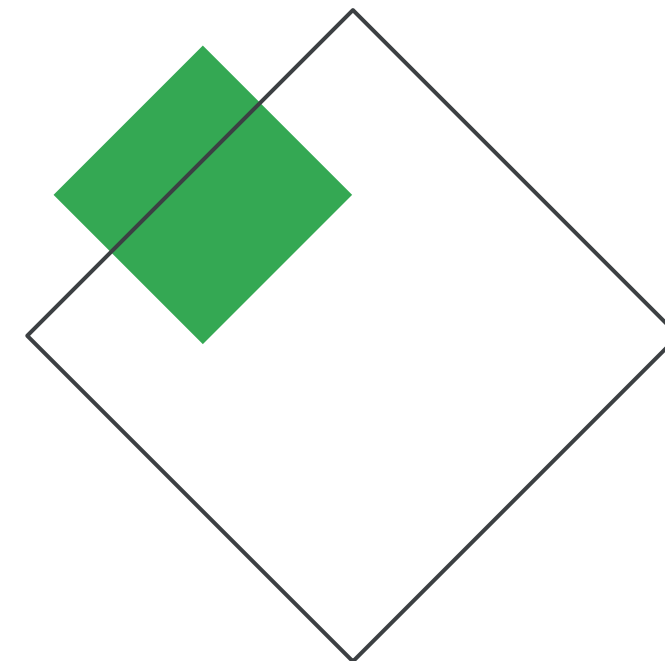
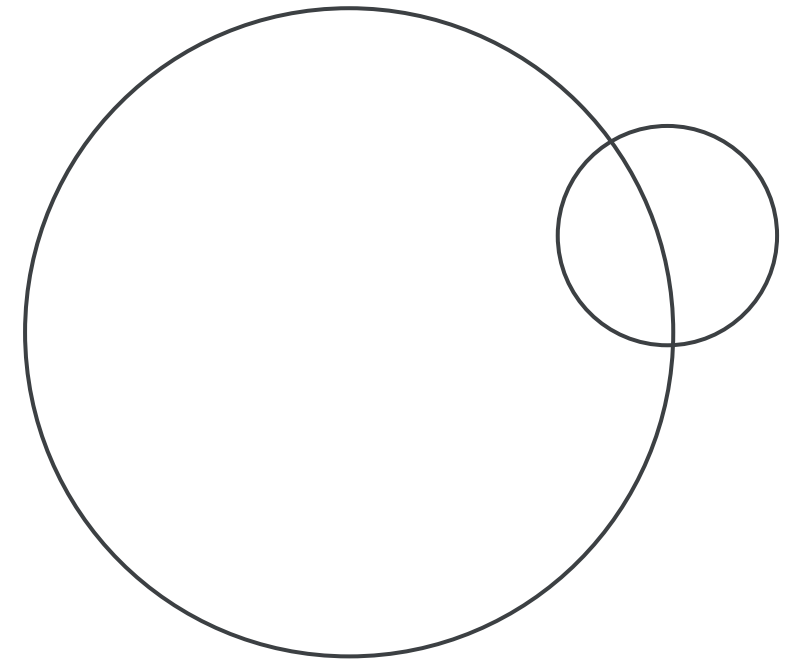
What type of cybersecurity threat are they facing?



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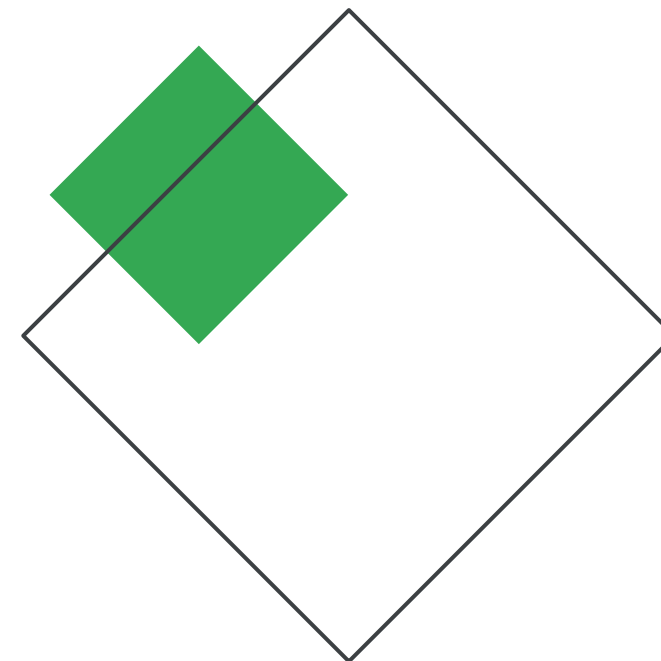
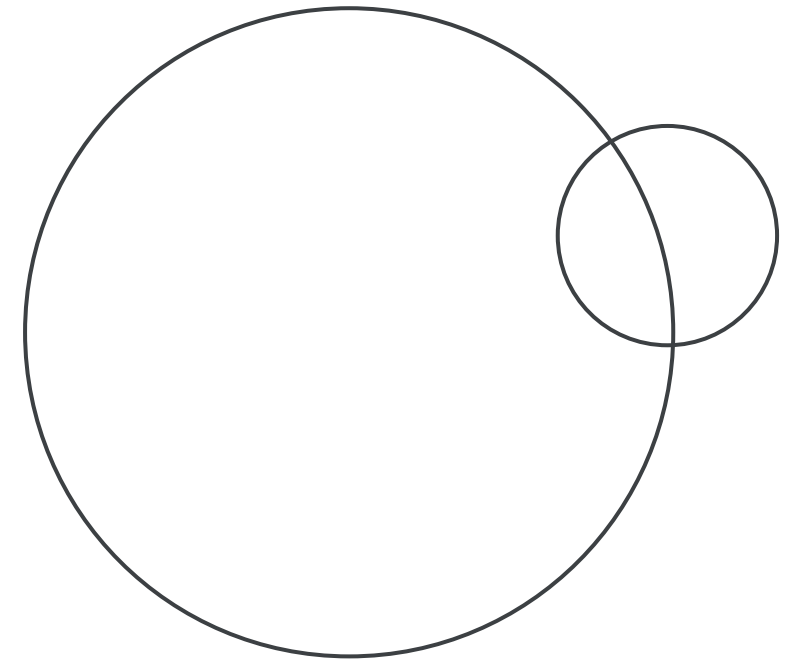
Answer: A malware or virus attack



Example 2

A colleague is searching for an application online. They find the software on an unfamiliar website, download, and run it. Their files then become locked and unusable, with an accompanying message demanding a payout to release the information.

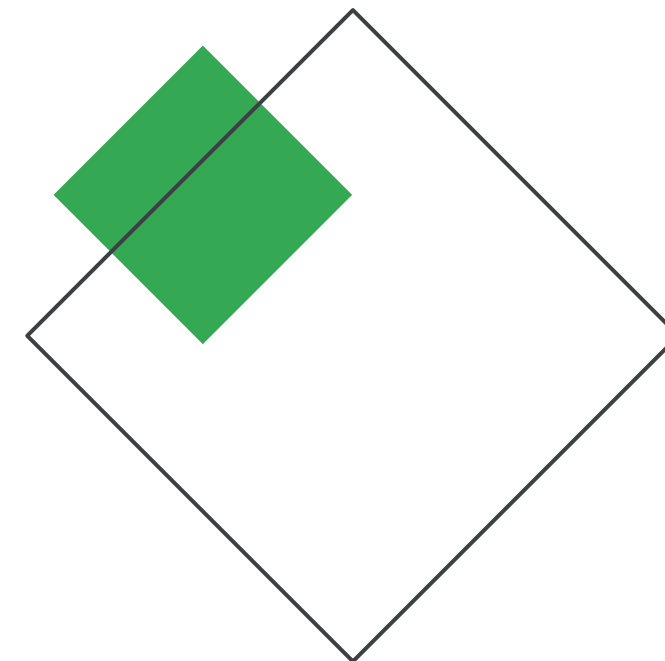
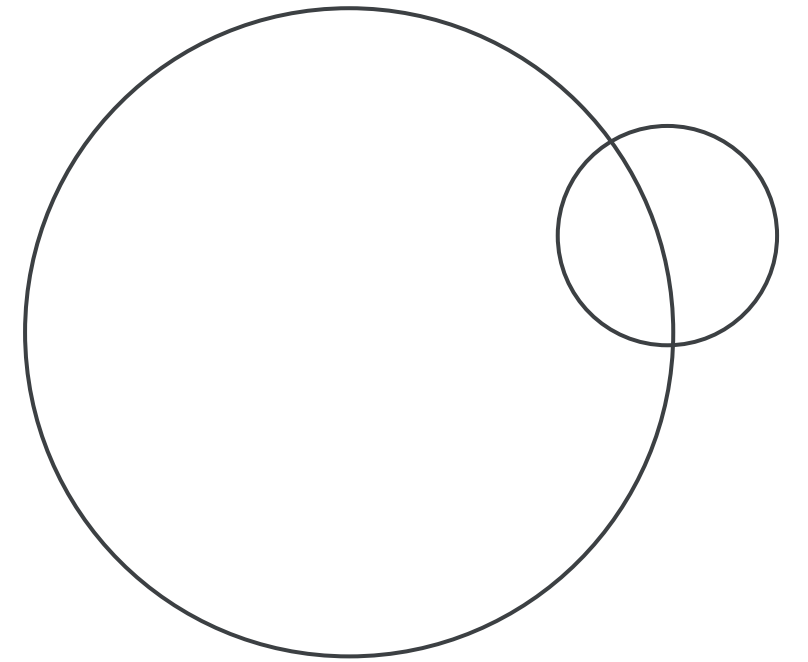
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Answer: A ransomware attack



Module 5

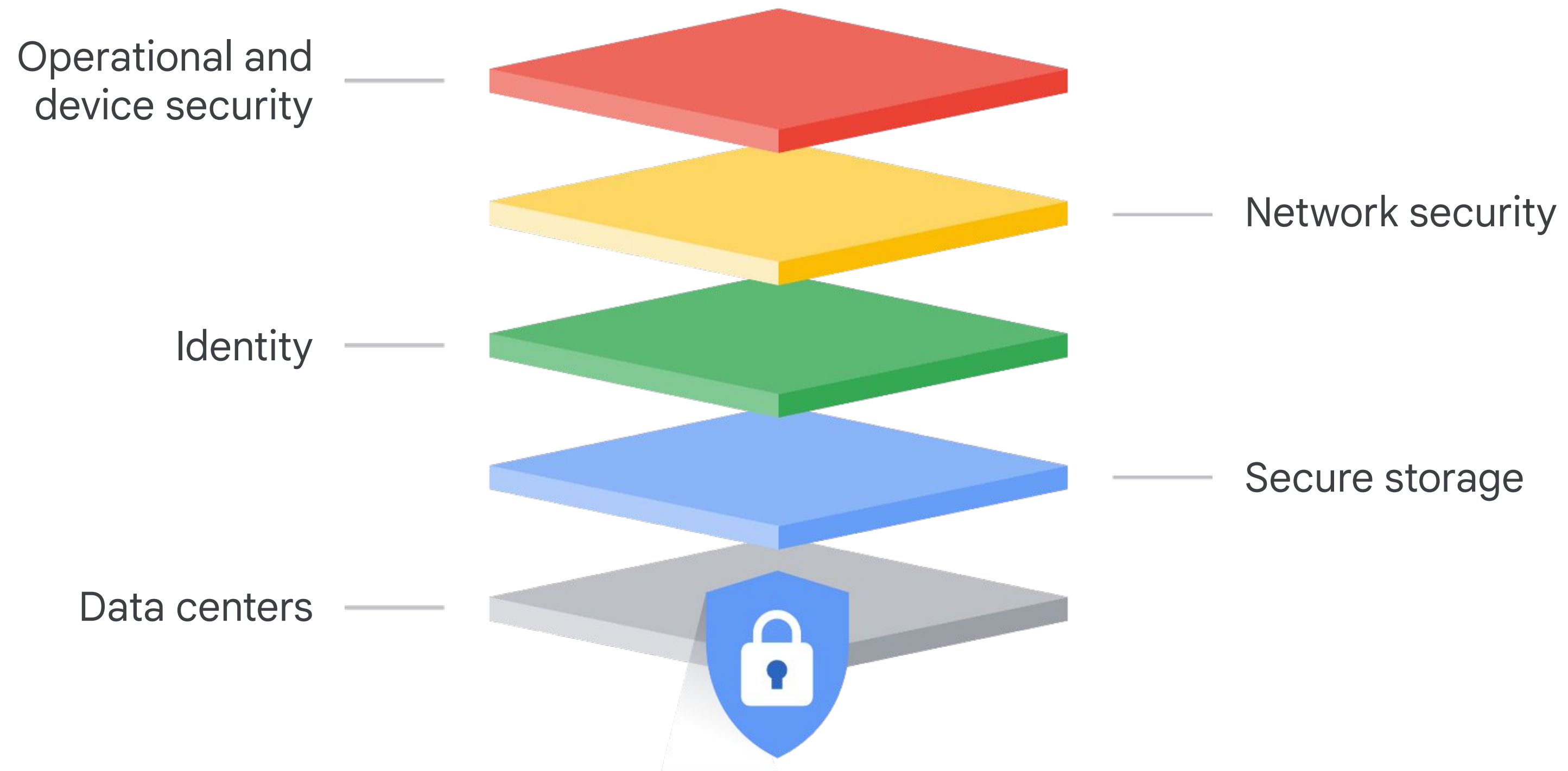
Trust and Security with Google Cloud



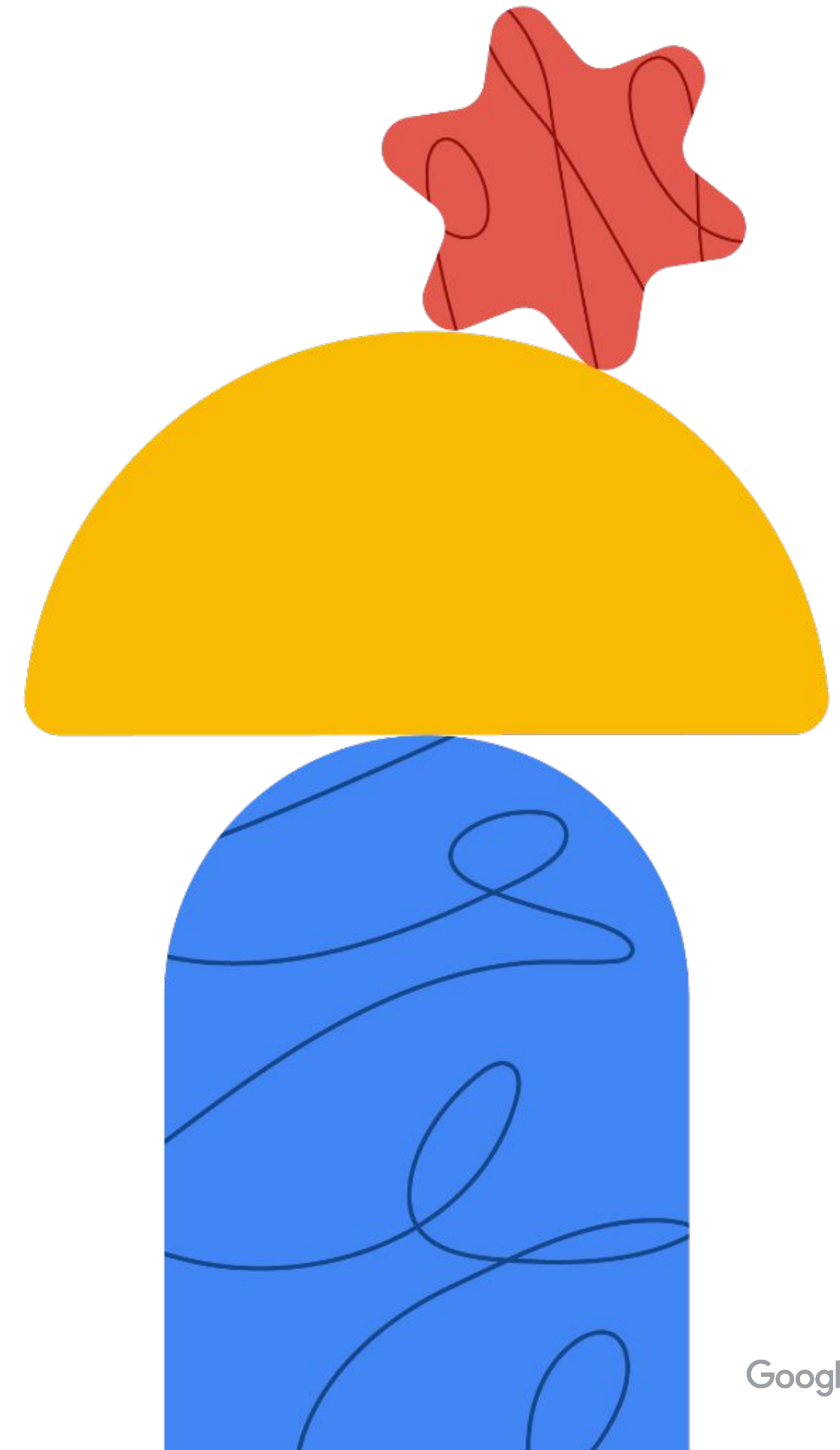
Lessons

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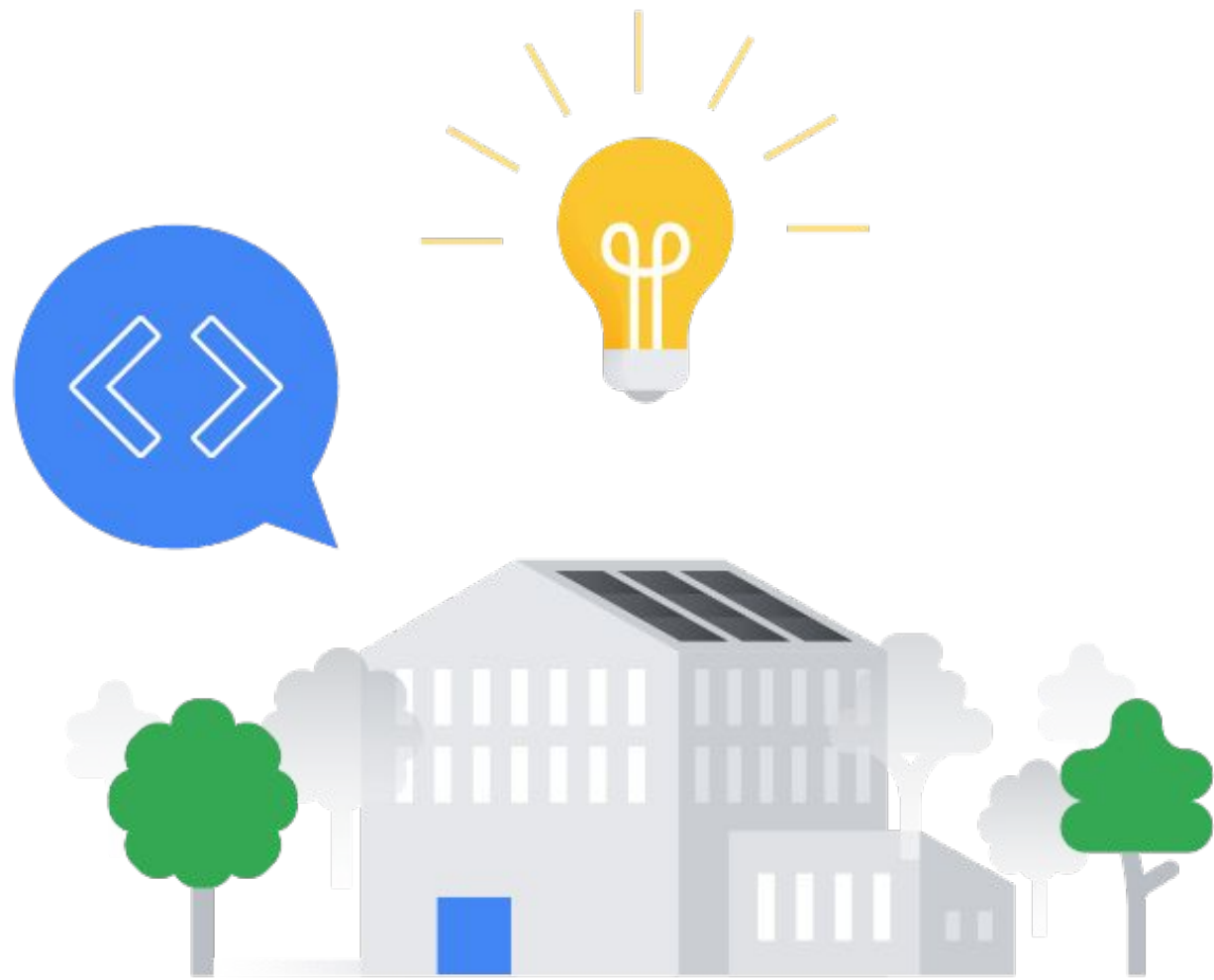
Google's multilayered security strategy



Data centers



Google operates over 30 state-of-the-art data centers worldwide



- ✓ Purpose-built servers
- ✓ Advanced networking solutions
- ✓ Custom security hardware and software

Zero-trust architecture

Our custom hardware and software are purpose-built with features such as:

- ✓ Tamper-evident hardware
- ✓ Secure boot
- ✓ Hardware-based encryption

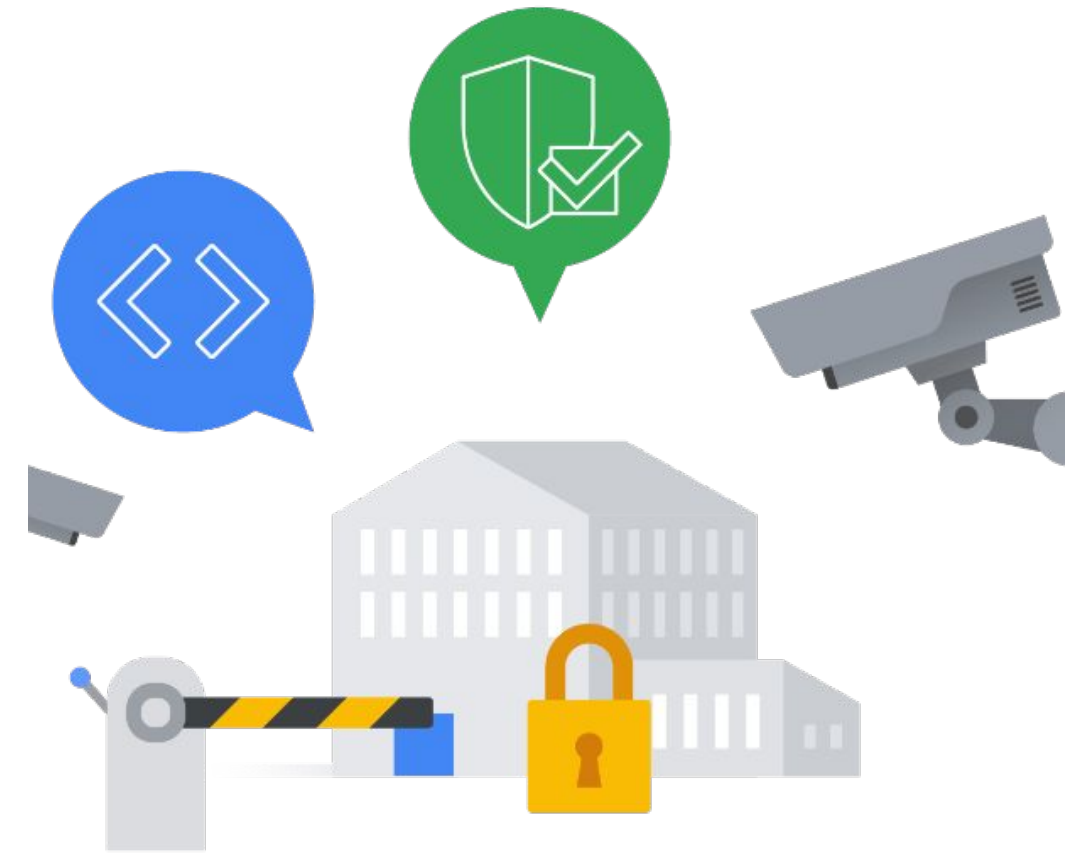


Physical security



Robust access control measures

Biometric authentication in place



Security by default

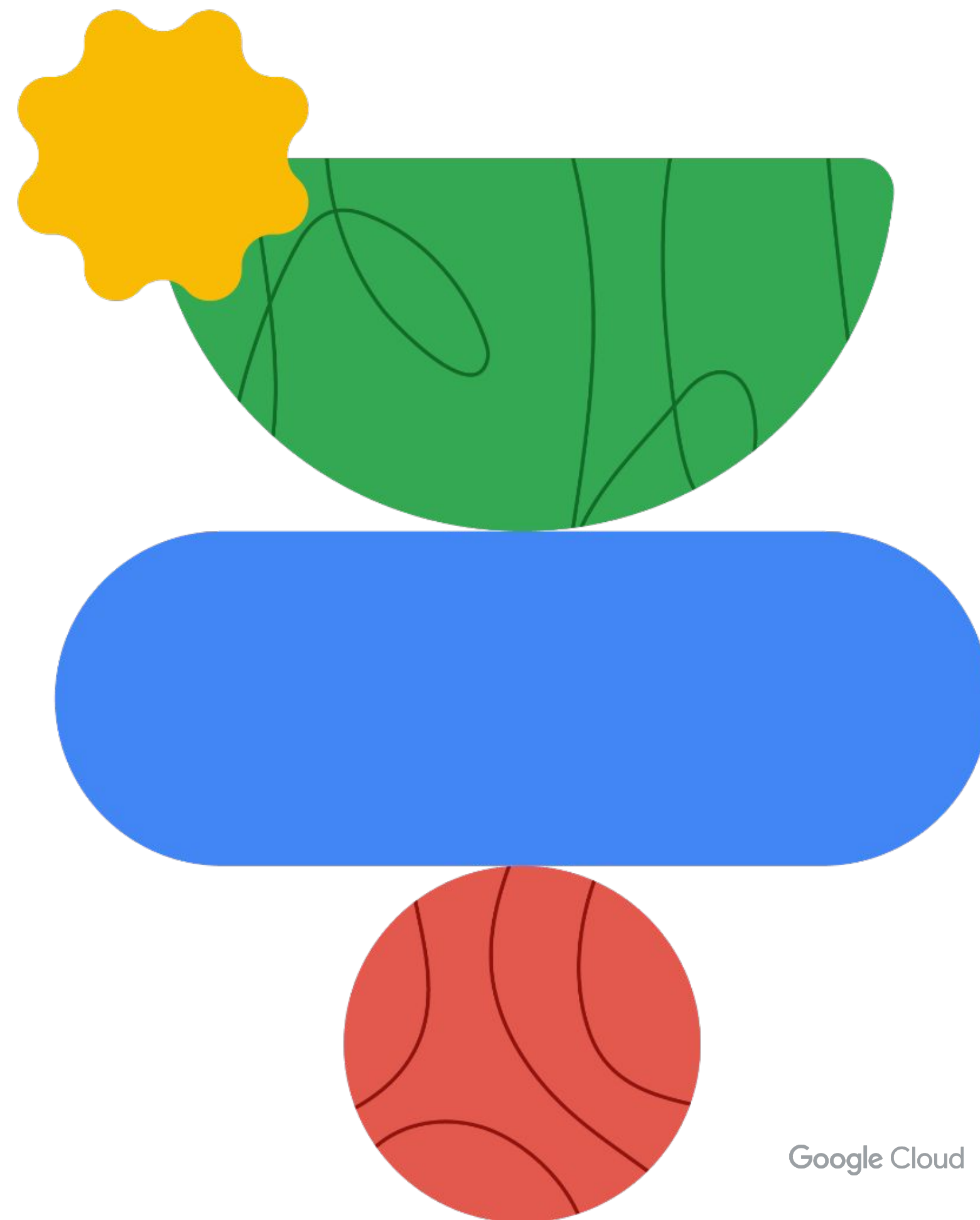
Designed with security in mind

Cyber resilience

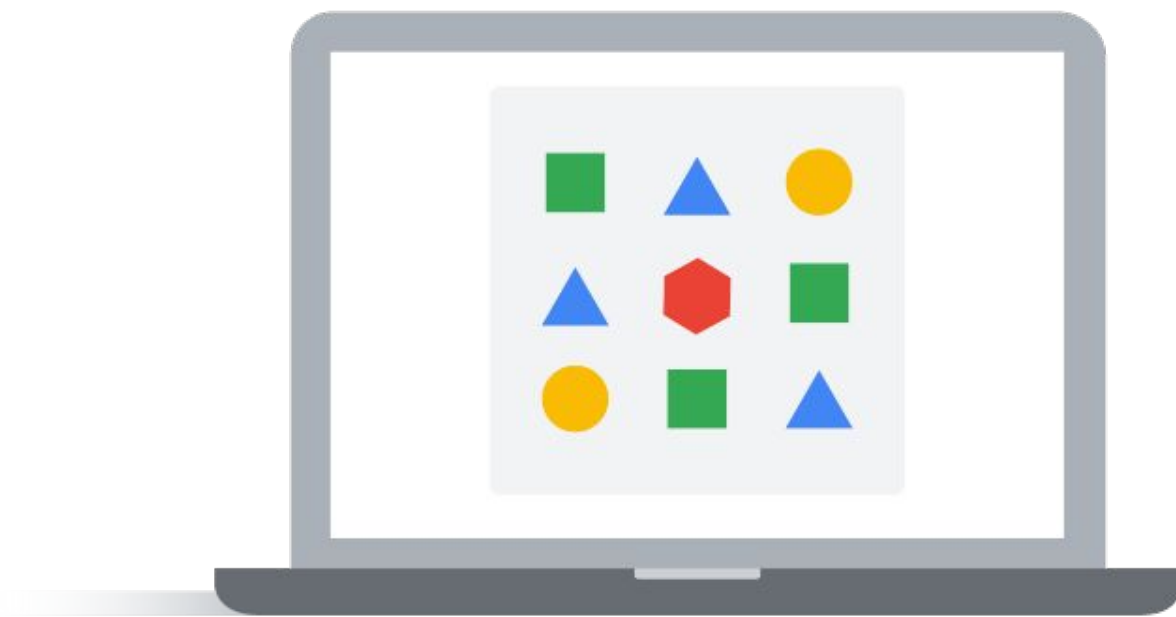


Our data centers are equipped to withstand and recover from potential security incidents, and ensure the continuity and integrity of data.

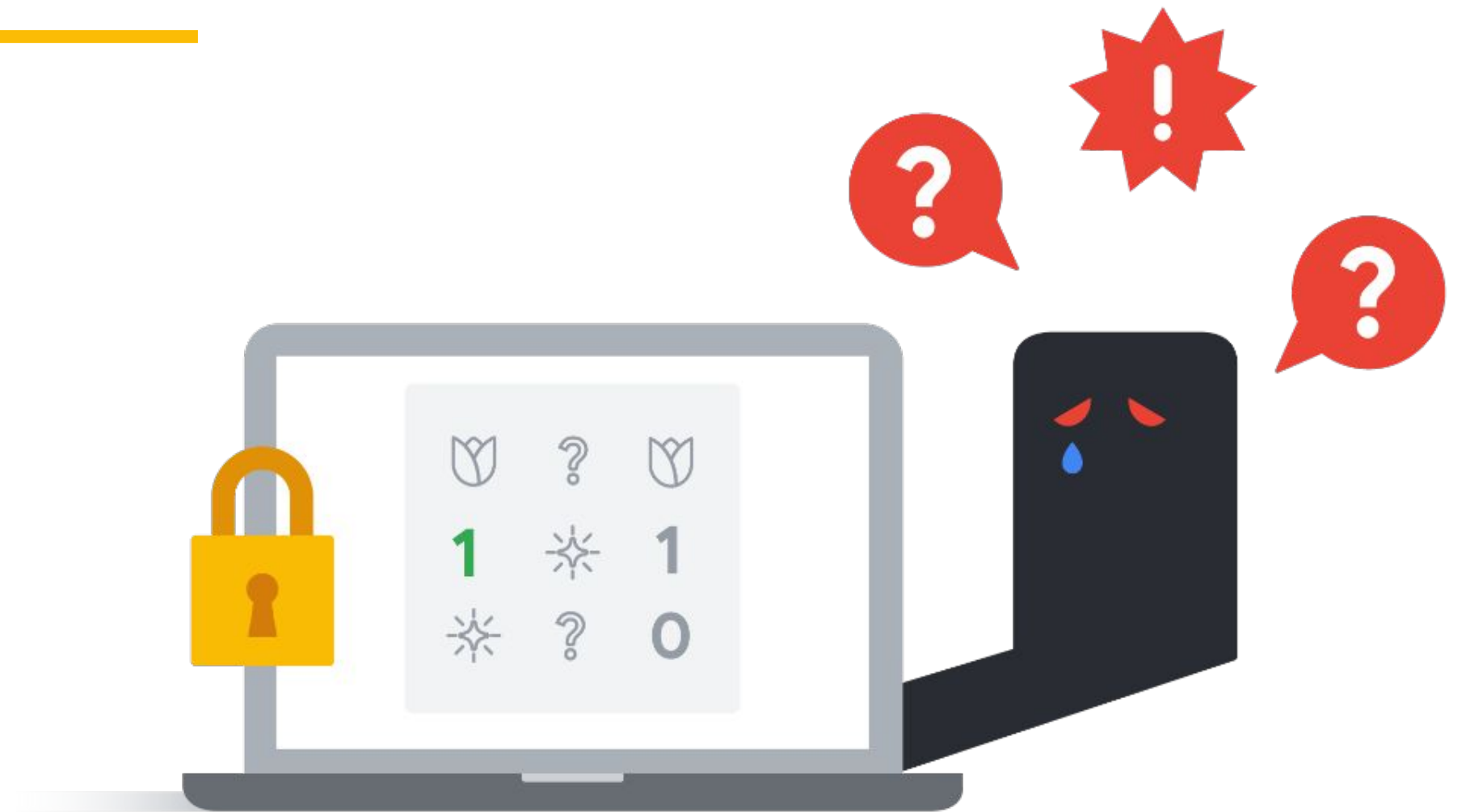
Secure storage



Encryption protects data at rest

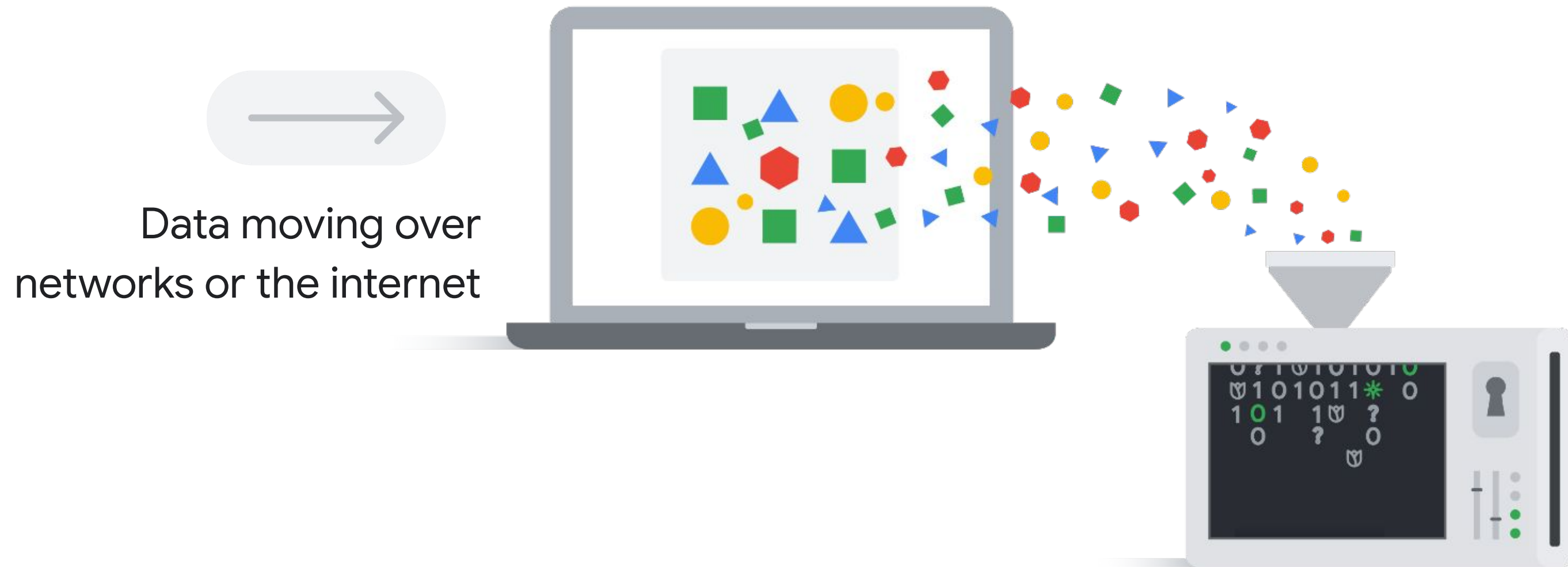


When data is at rest, it's stored on physical devices like computers or servers.



Even if someone gains physical access to the device, they won't be able to decipher the data without the encryption key.

Encryption protects data in transit

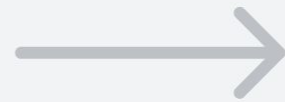


Encryption shields data from interception by cybercriminals or unauthorized parties.

Encryption protects data in use

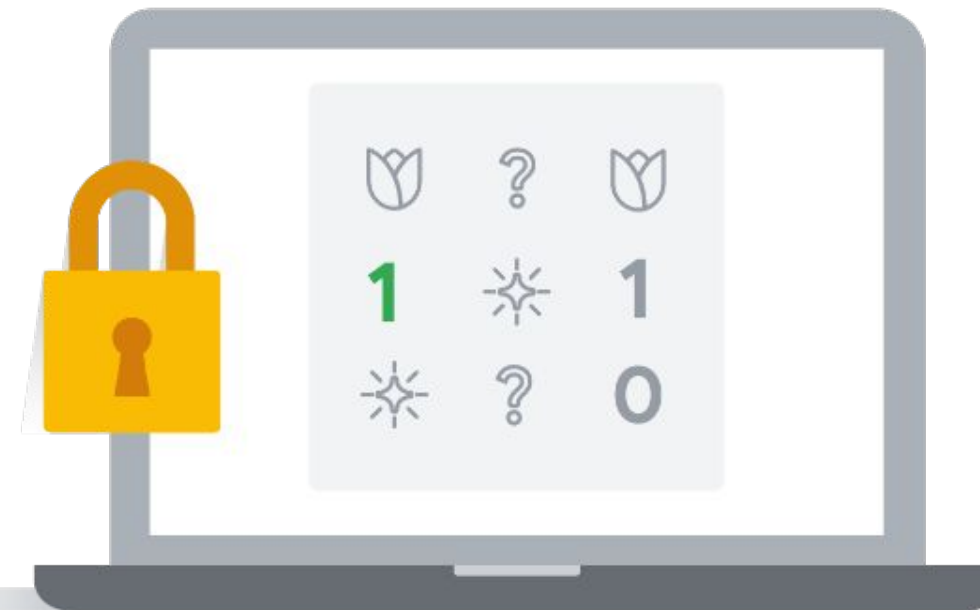
Data in use

Data being actively processed by a computer.

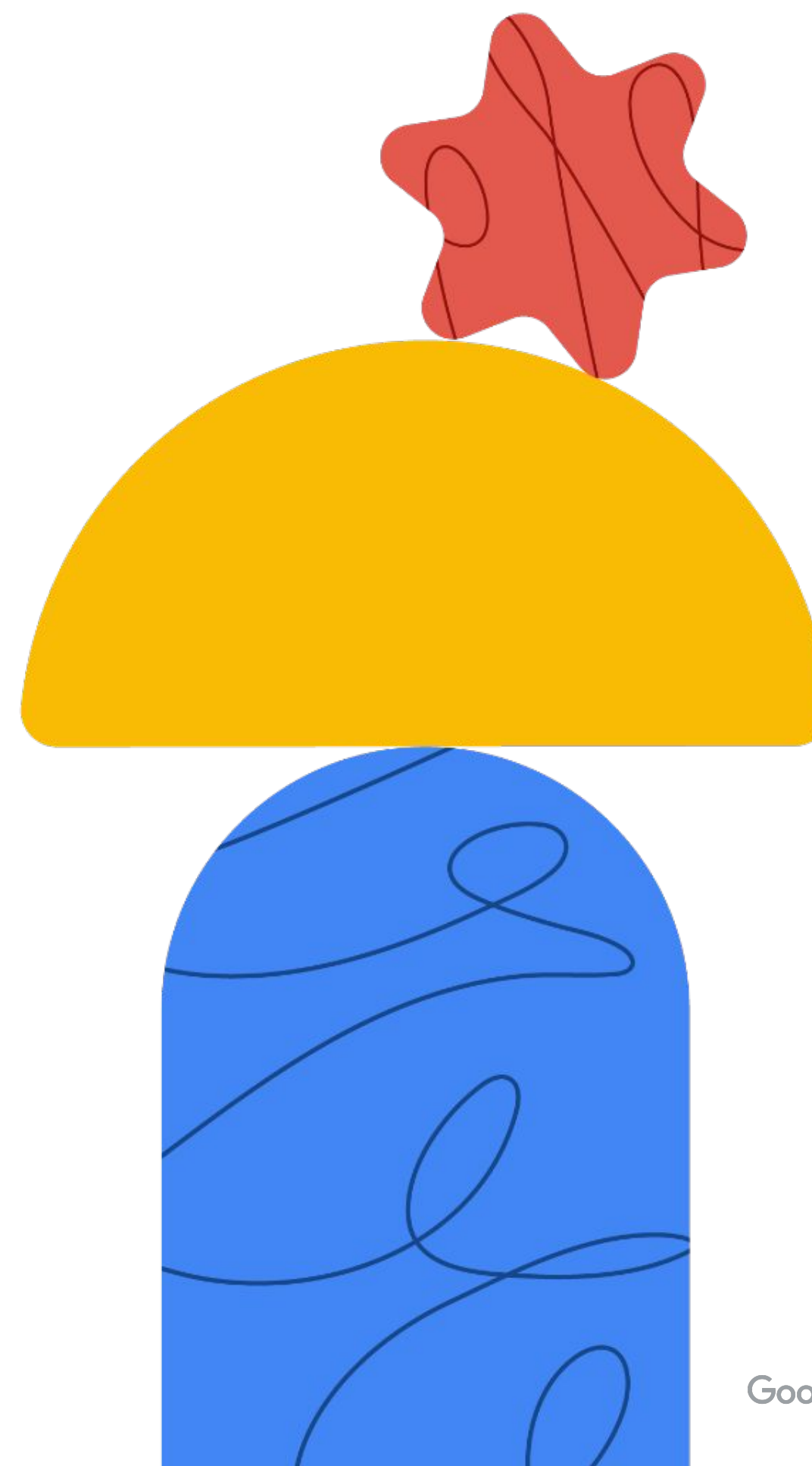


Memory encryption

Locks your data inside the computer's memory,



Identity



The three A's of cloud identity management



Authentication

Authorization

Auditing

Used to:

- ✓ Ensure secure access
- ✓ Manage user privileges
- ✓ Monitor resource usage

Authentication

Authentication verifies the identity of users or systems that seek access through unique credentials.



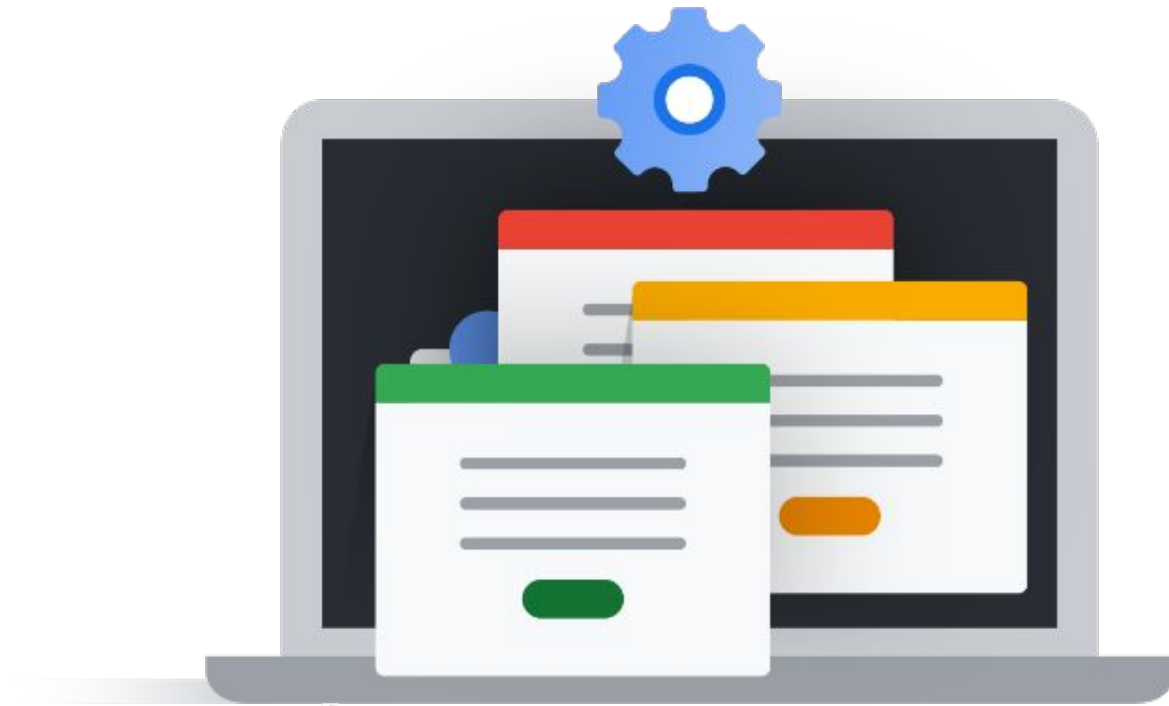
Authorization

Authorization determines what a user is allowed to do within the system.



Admin

Create and remove user accounts

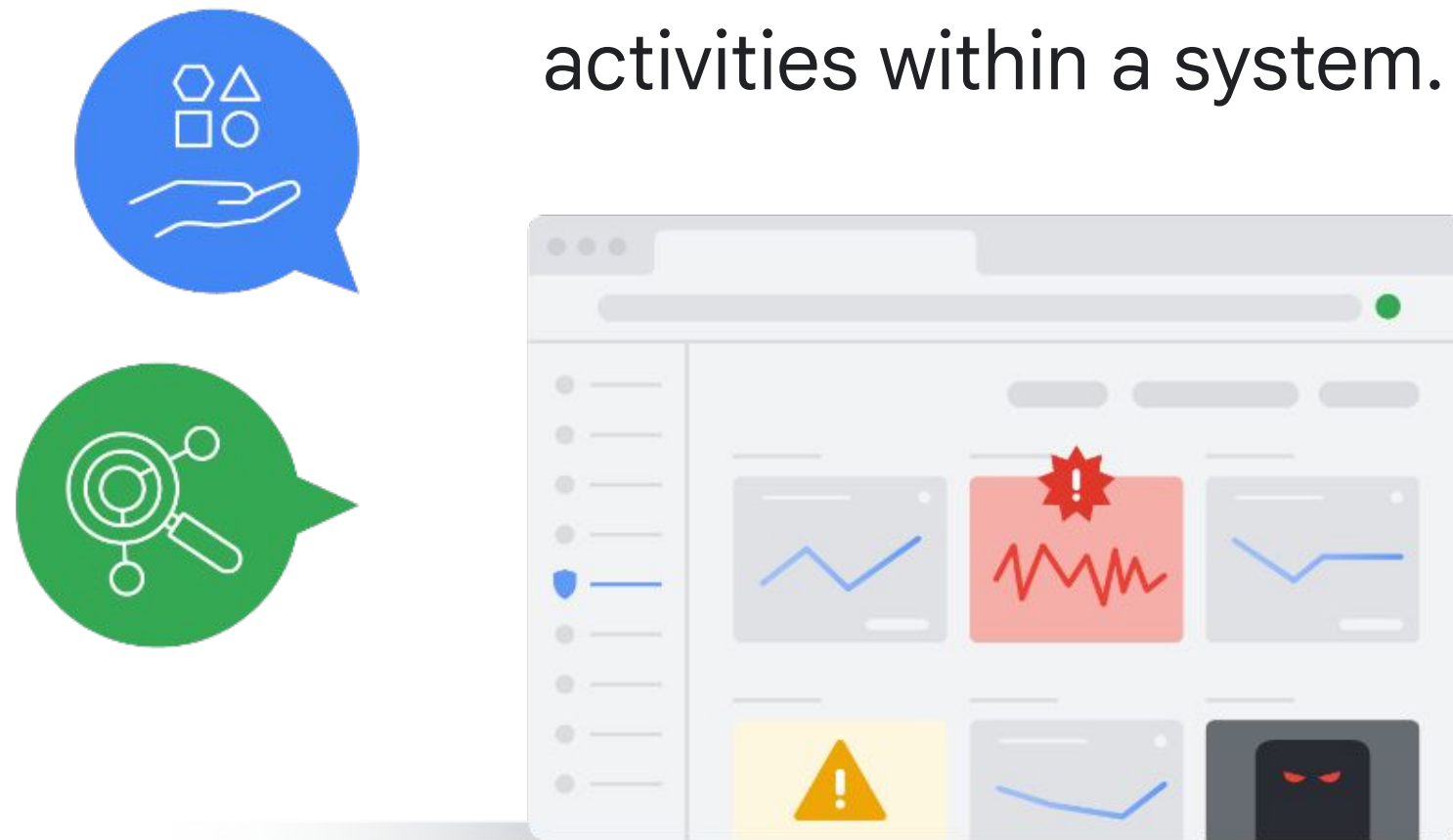


Standard user

View a list of users

Auditing (or accounting)

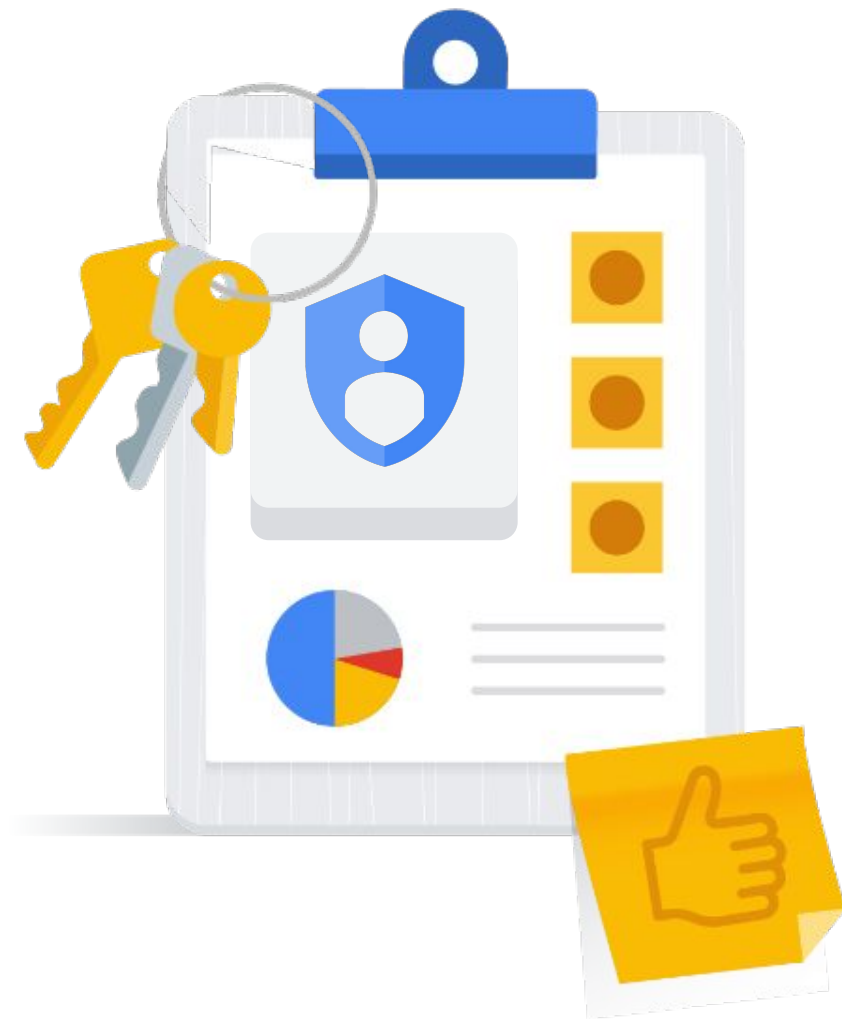
Auditing plays a critical role in monitoring and tracking user activities within a system.



It provides a comprehensive record of actions taken on a system, which is helpful during:

- ✓ Security incident investigations
- ✓ Compliance tracking
- ✓ System performance evaluation

Use Identity and Access Management (IAM) for granular access control



Identity and Access
Management (IAM)

- ✓ Create and manage user accounts.
- ✓ Assign roles to users.
- ✓ Grant and revoke permissions to resources.
- ✓ Audit user activity.
- ✓ Monitor your security position.

Quiz

Question

Which aspect of cloud identity management verifies the identity of users or systems?

- A. Authorization
- B. Accounting
- C. Auditing
- D. Authentication

Quiz

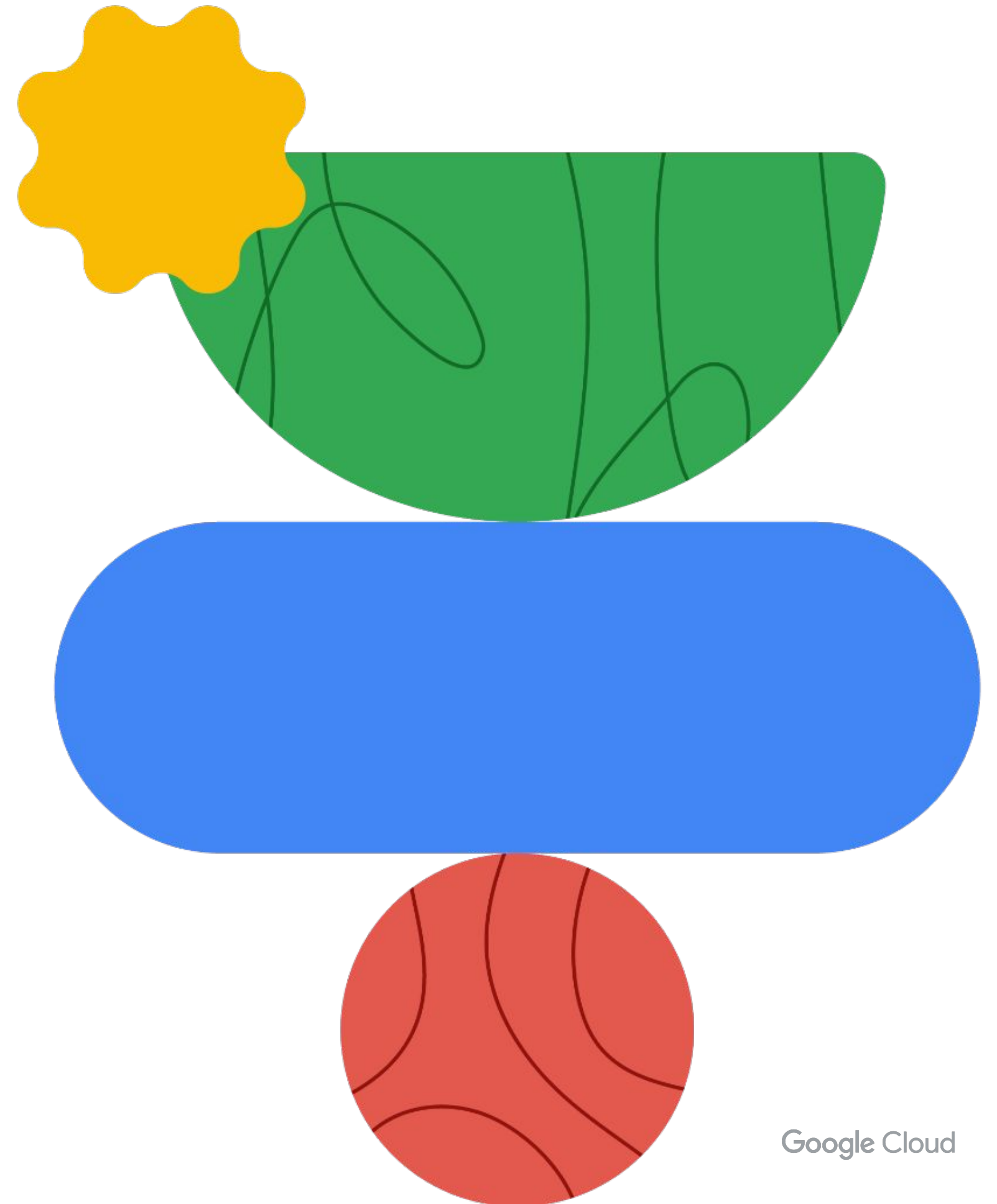
Answer

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Network security

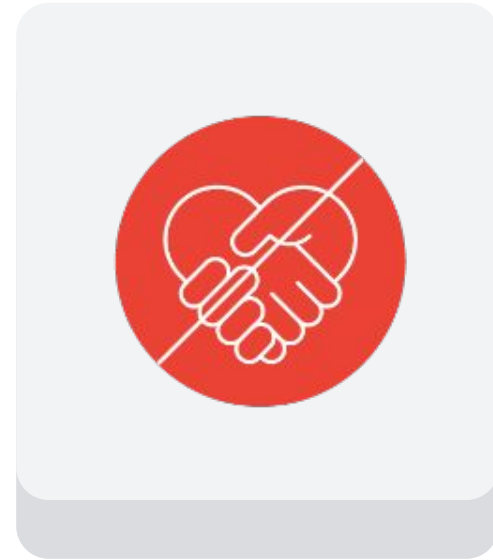


Strategies to secure your organization's network

- Embrace the power of zero-trust networks.
- Secure your connections to on-premises and multi-cloud environments.
- Protect your perimeter with Google Cloud's powerful tools.
- Stay ahead with a web application firewall.
- Automate infrastructure provisioning for enhanced security.

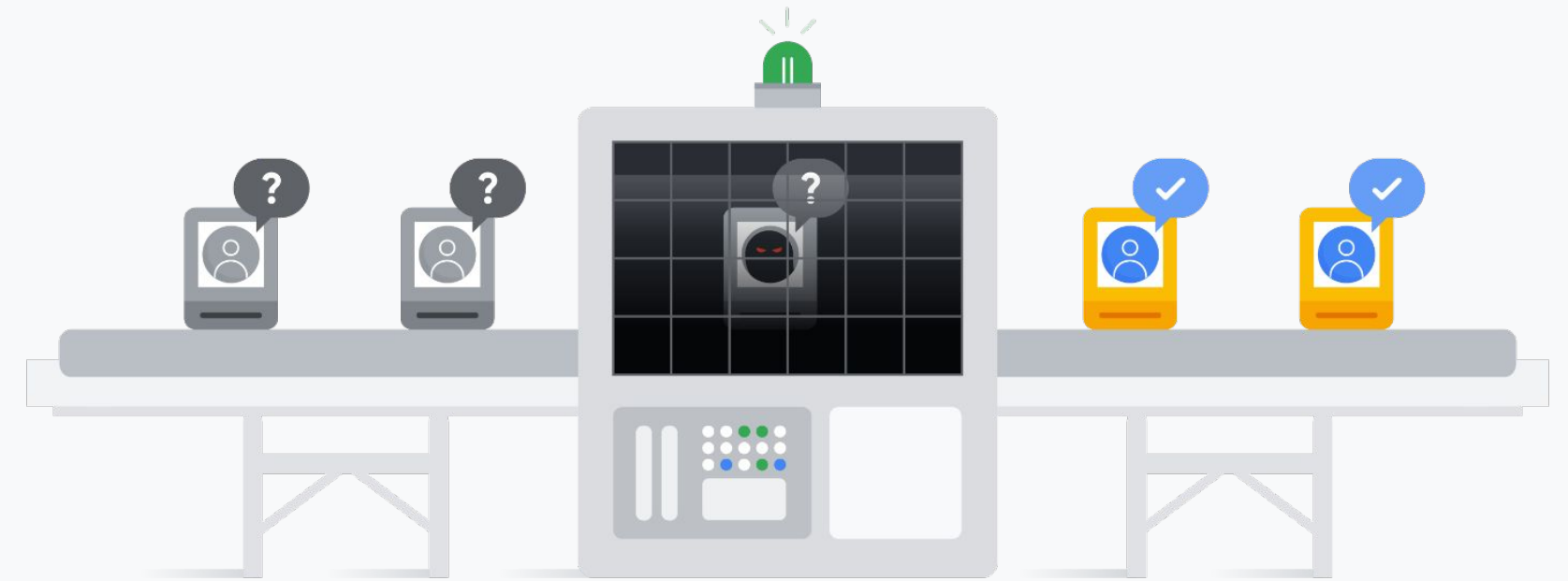


Embrace the power of zero-trust networks



BeyondCorp
Enterprise

Use to implement a
zero-trust security
model



Every access request is thoroughly
verified, and both the user's identity
and context are considered.

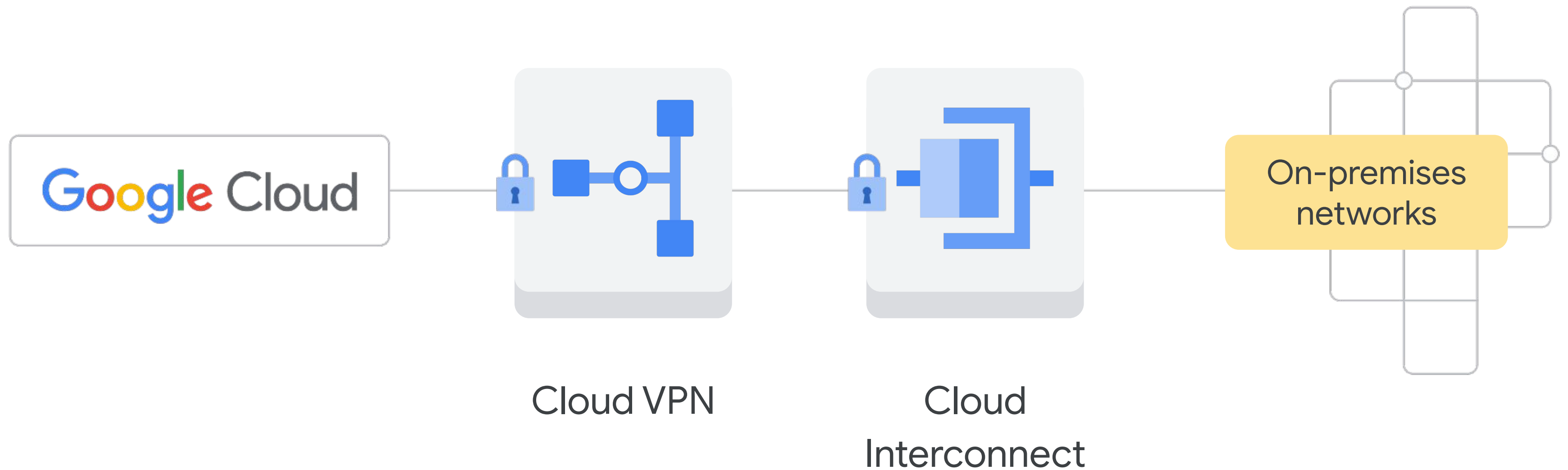
Secure your connections to on-premises and multi-cloud environments



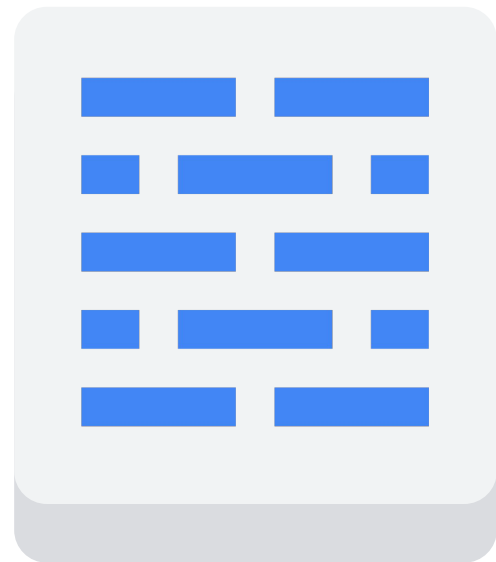
On-premises

Multi-cloud

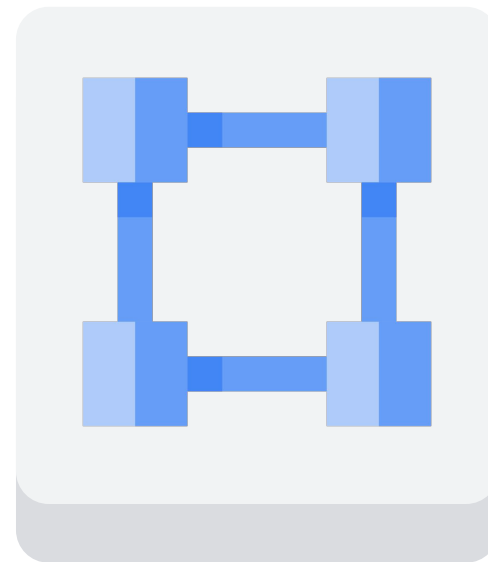
Google Cloud provides private access methods



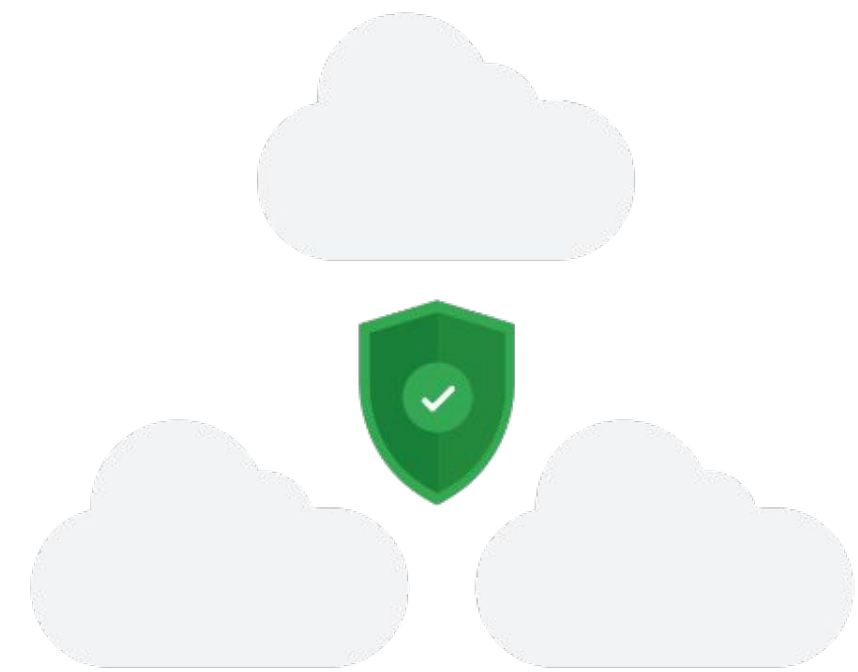
Protect your perimeter with Google Cloud's powerful tools



Firewall

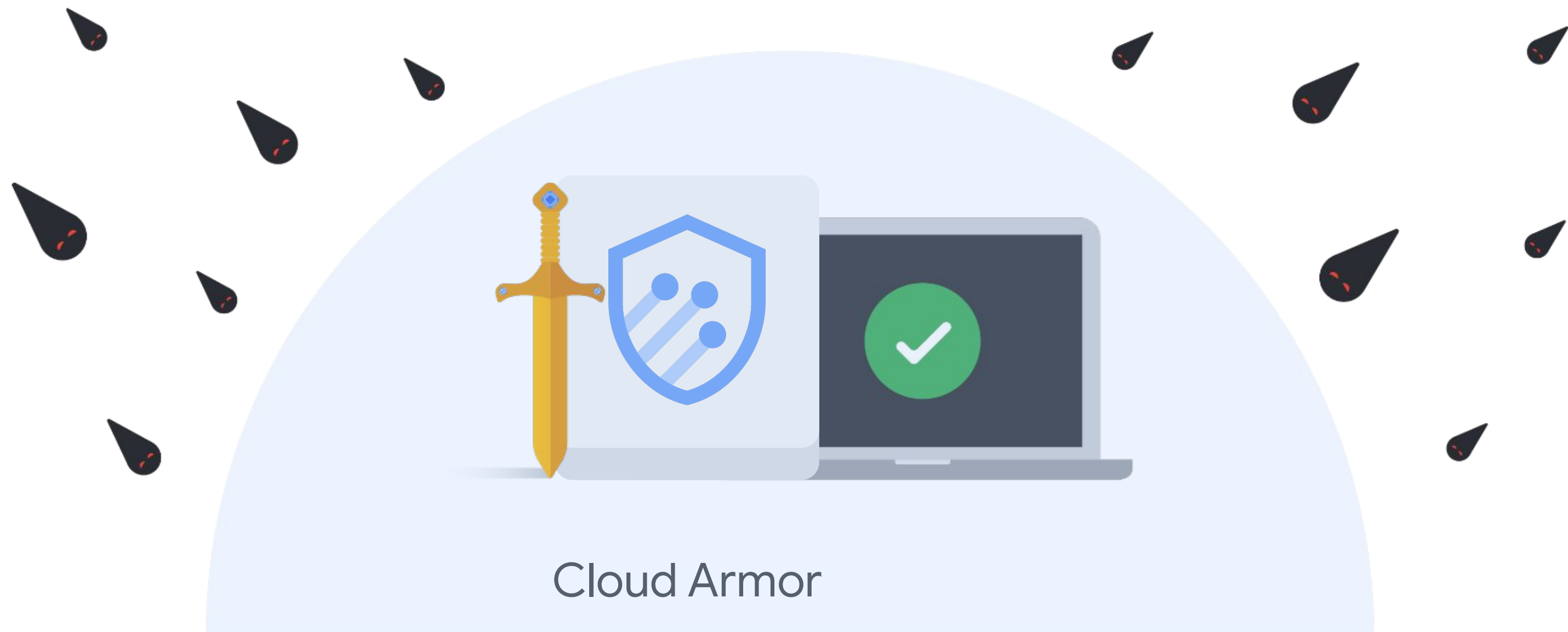


Virtual Private
Cloud

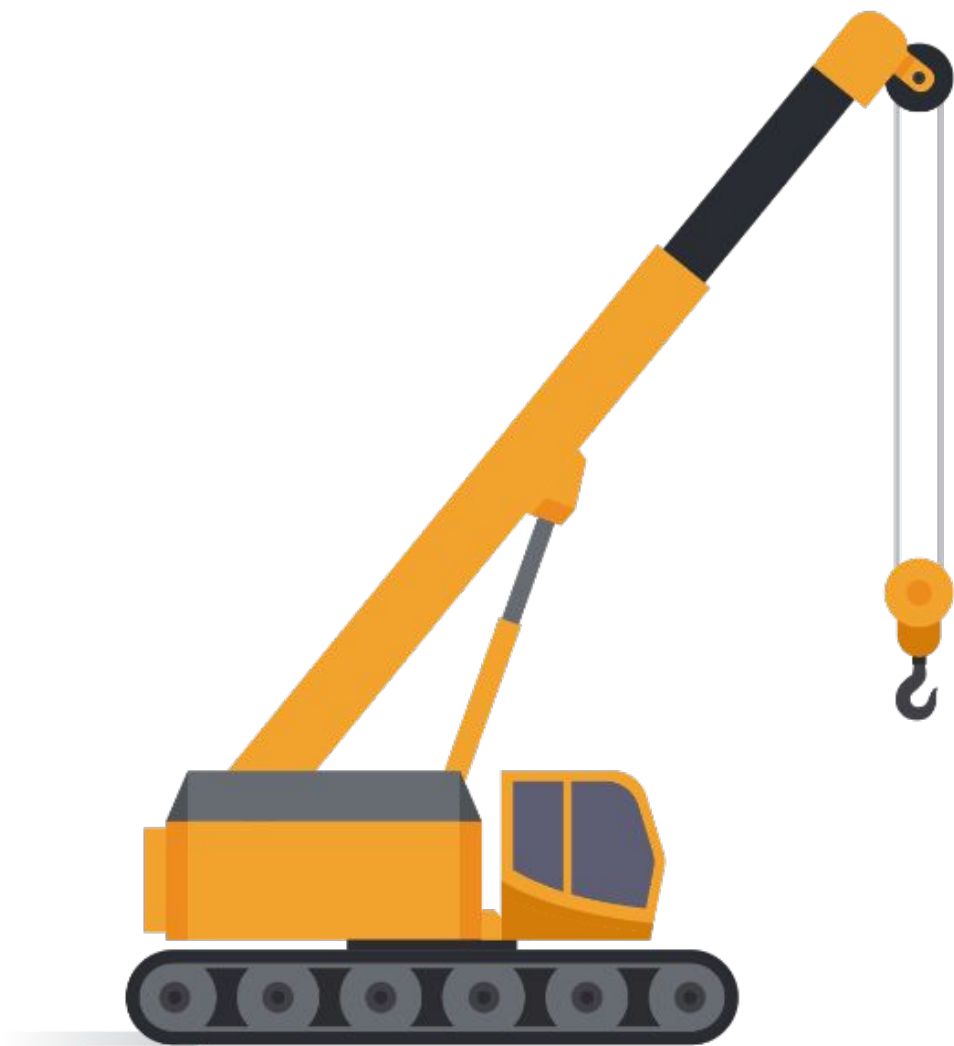


Google Cloud

Google Cloud Armor provides robust DDoS protection



Automate infrastructure provisioning for enhanced security

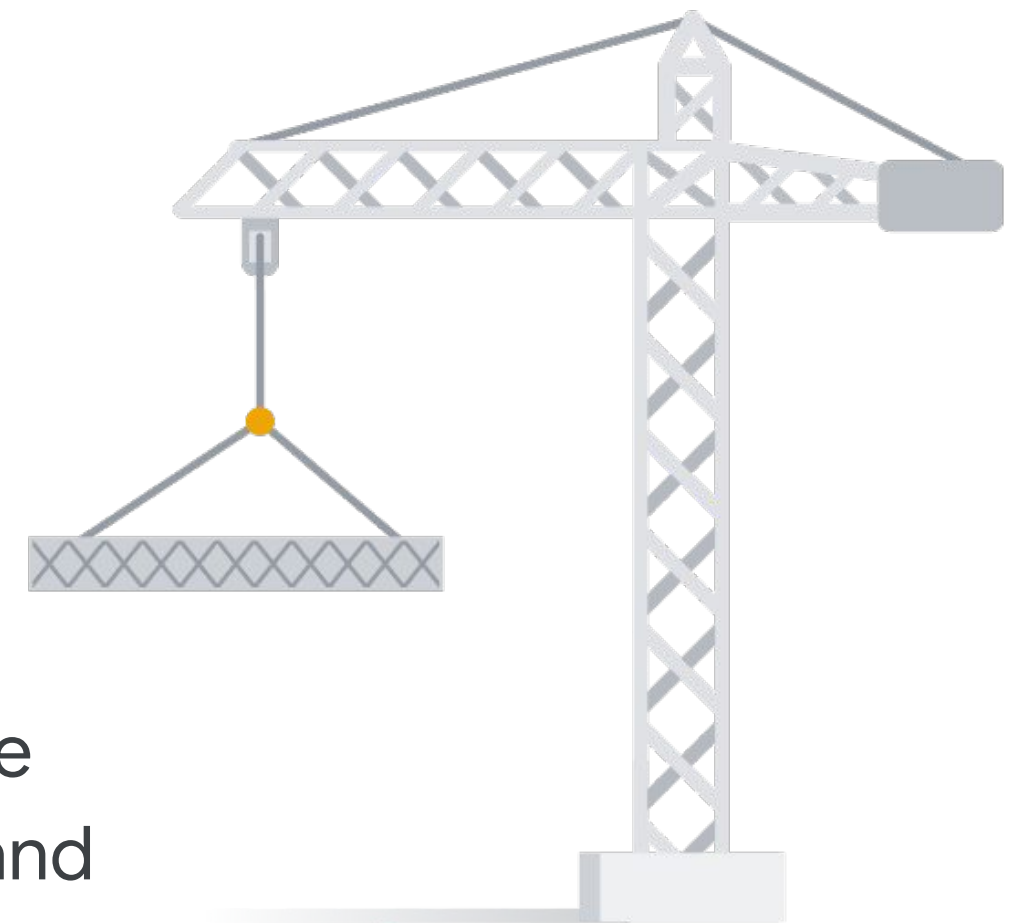


Terraform

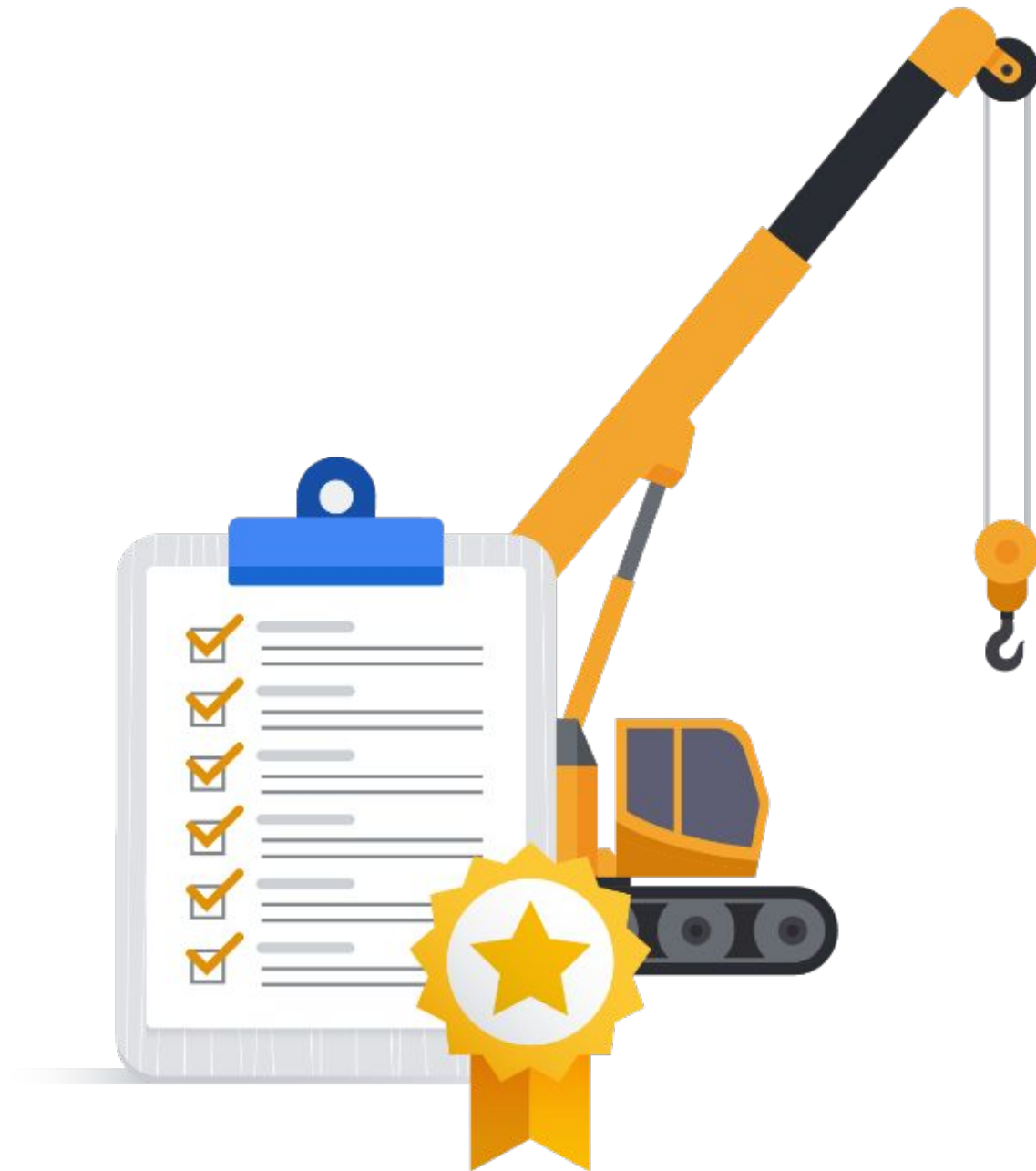
Jenkins

Cloud Build

These automation tools can handle all of the behind-the-scenes work to create a secure and reliable cloud environment.



There are benefits to using automation tools to enhance security



Your cloud environment becomes like a well-designed workspace.



When the environment is set up, it stays that way.

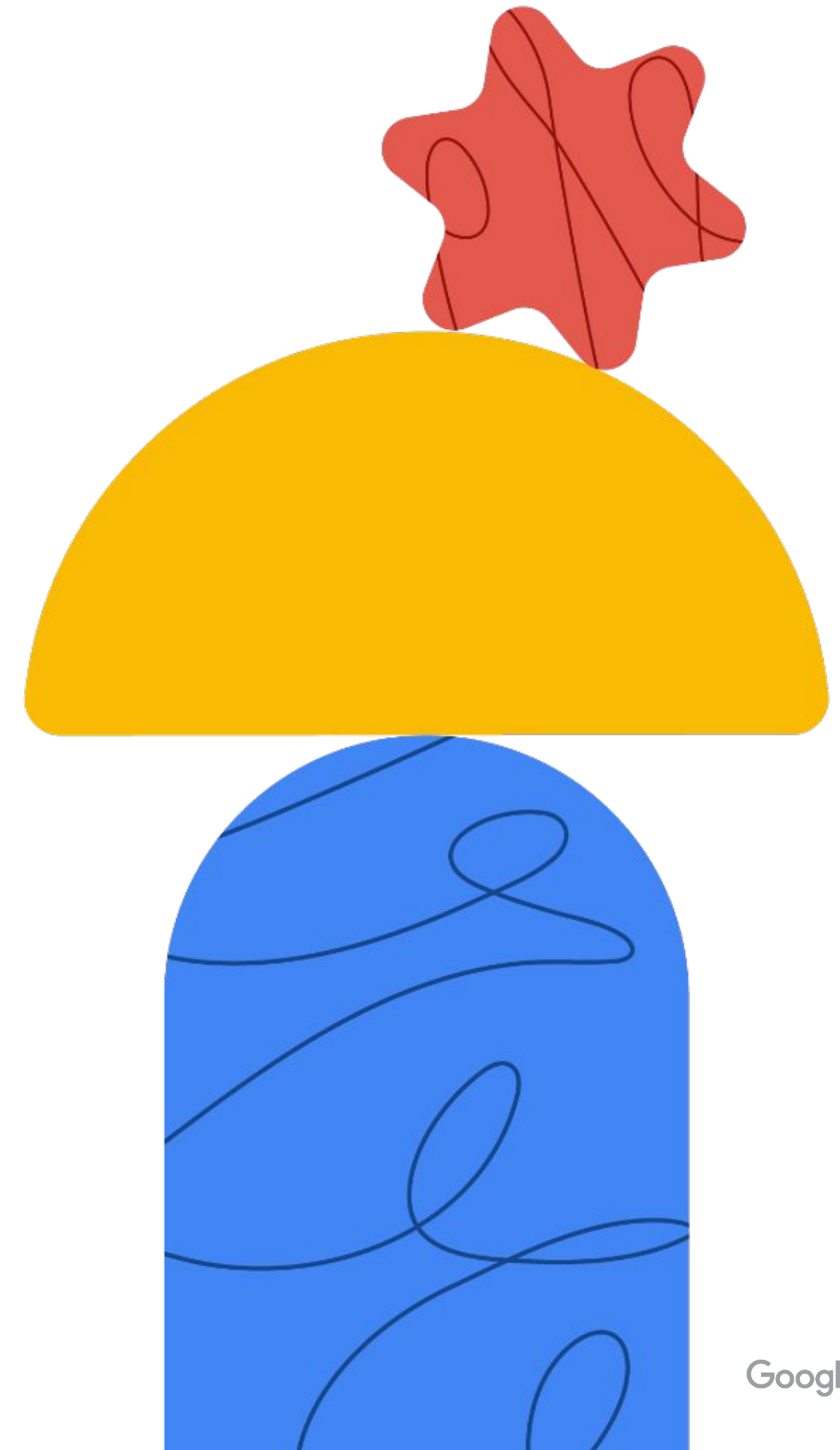


There are no unexpected changes or disruptions.

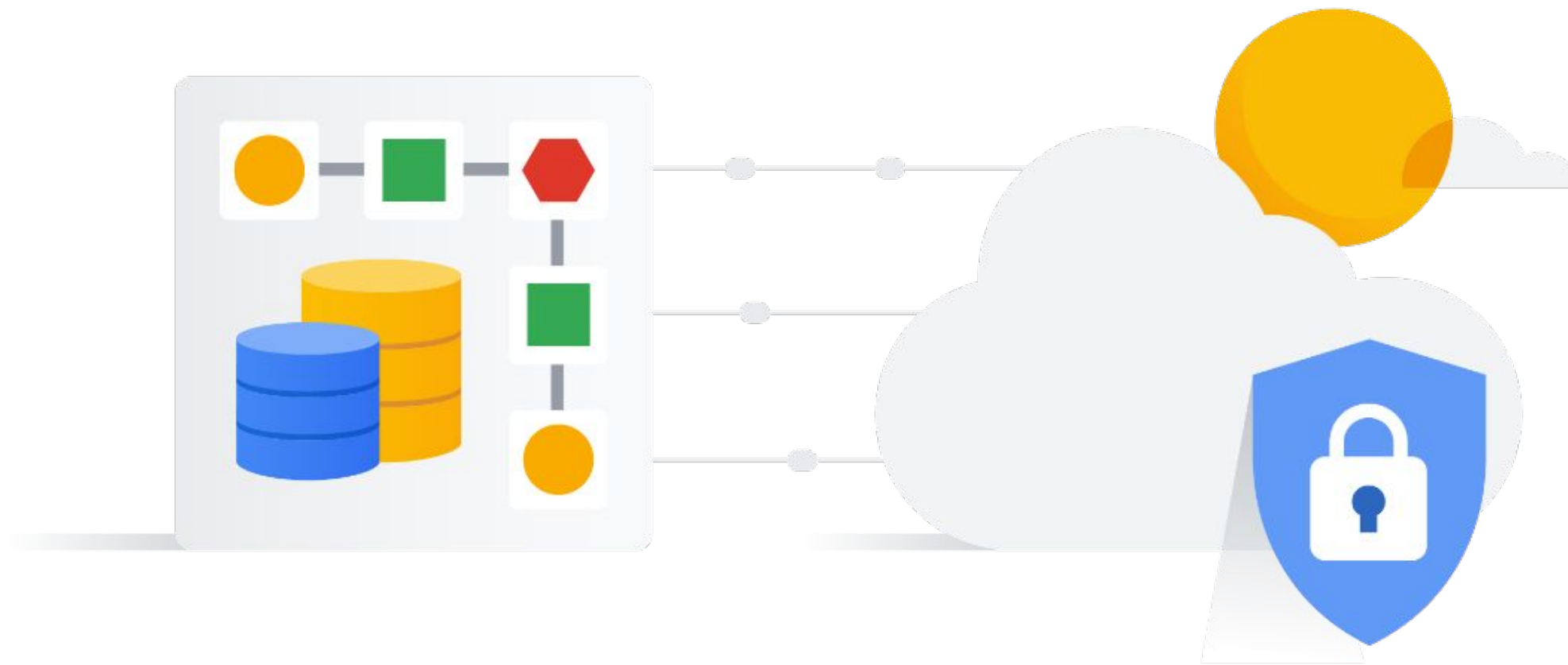


Tools can quickly identify and fix issues.

Security operations



SecOps → Security operations

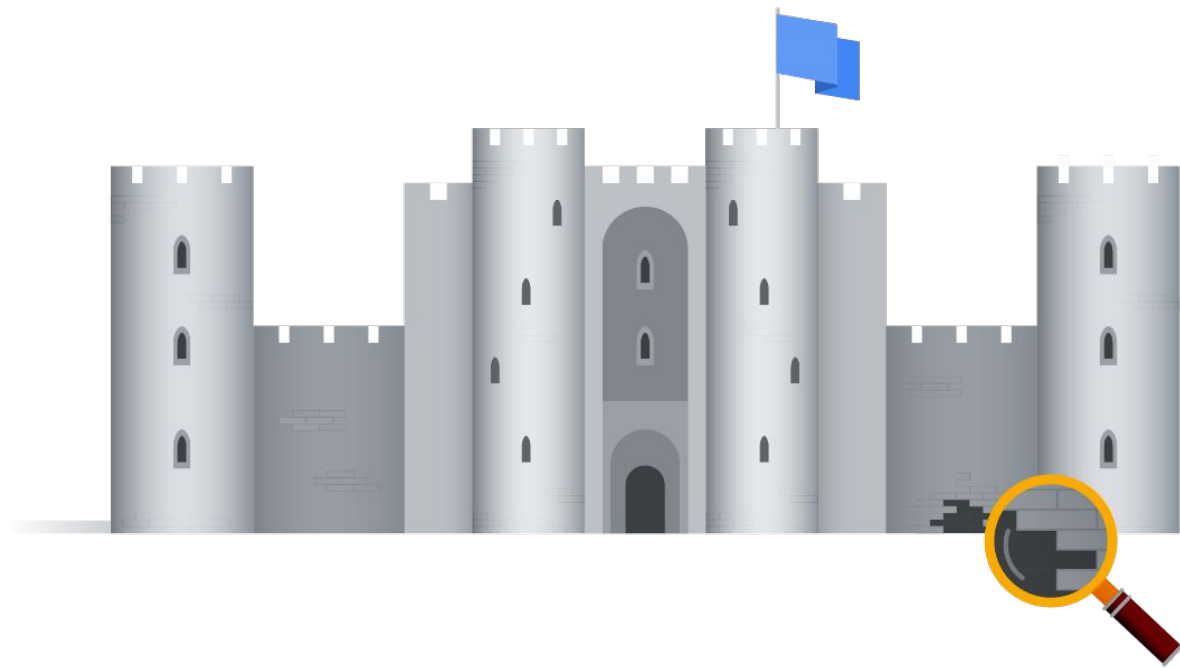


SecOps is all about protecting your organization's data and systems in the cloud.

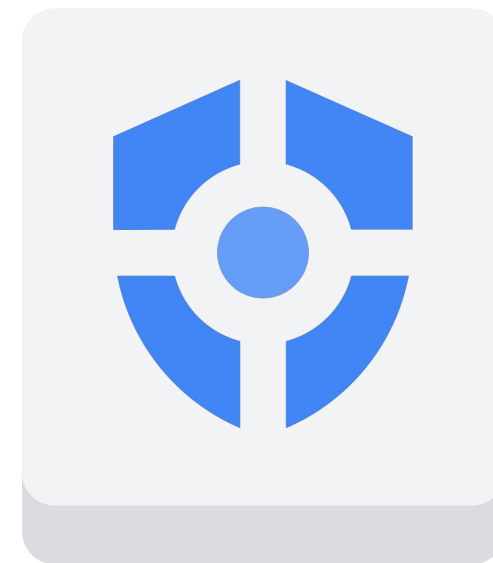
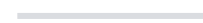
It helps reduce the risk of:

- ✓ Data breaches
- ✓ System outages
- ✓ Other security incidents

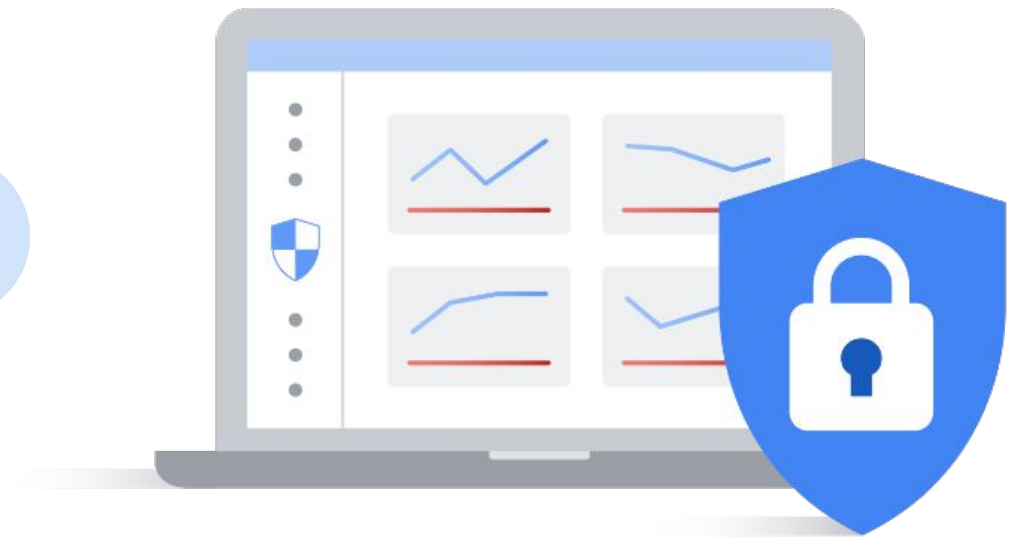
Vulnerability management



The process of identifying and fixing security vulnerabilities in cloud infrastructure and applications.

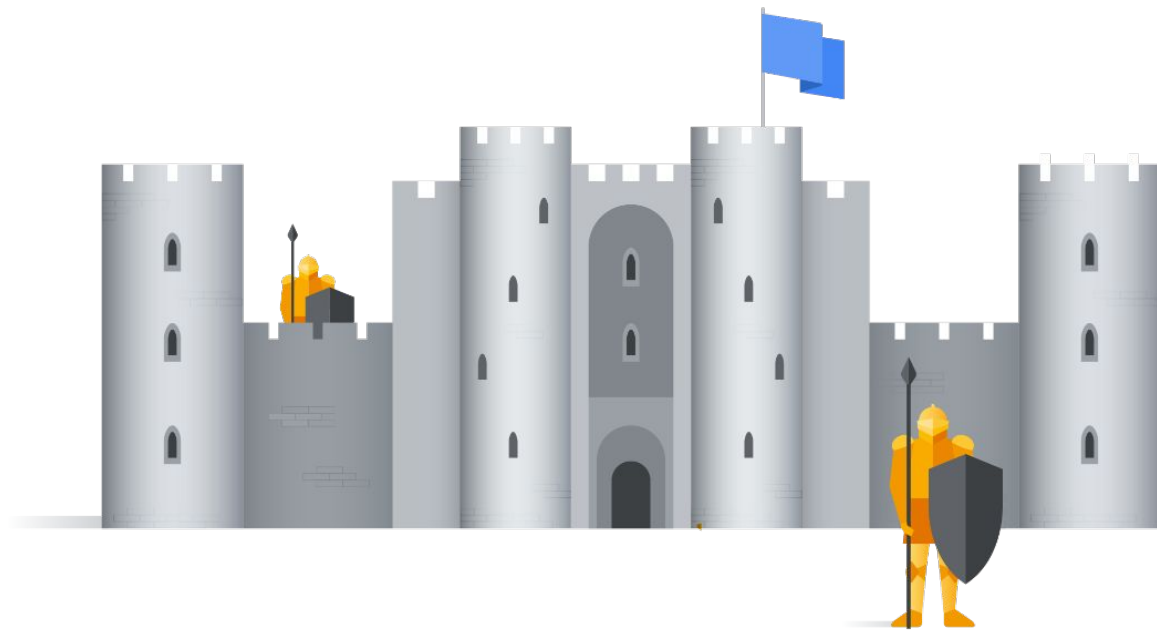


Security Command Center (SCC)

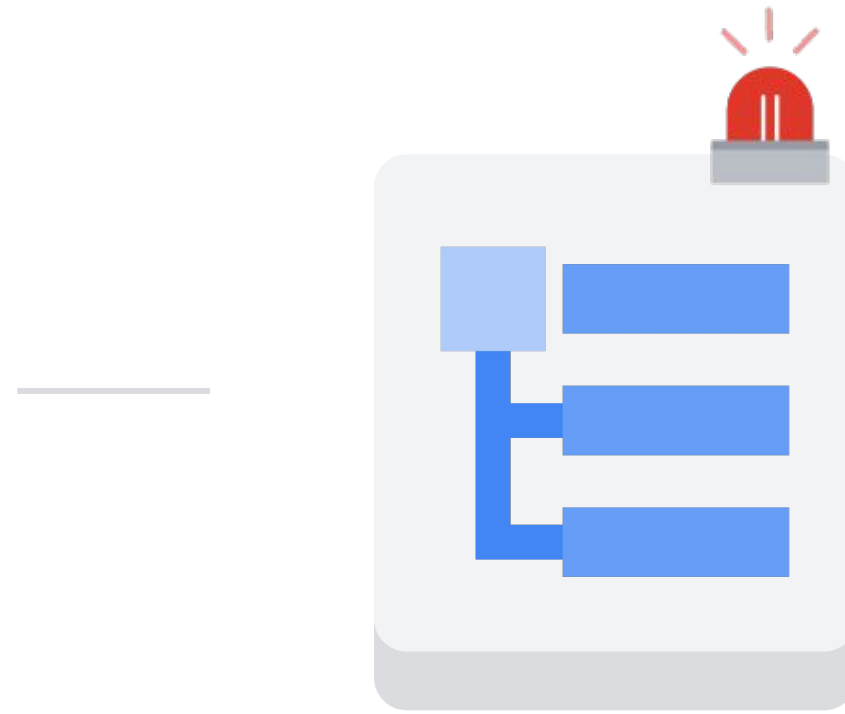


Provides a centralized view of your security posture.

Log management



It's like having a watchful eye on your castle grounds, looking out for any suspicious activity.

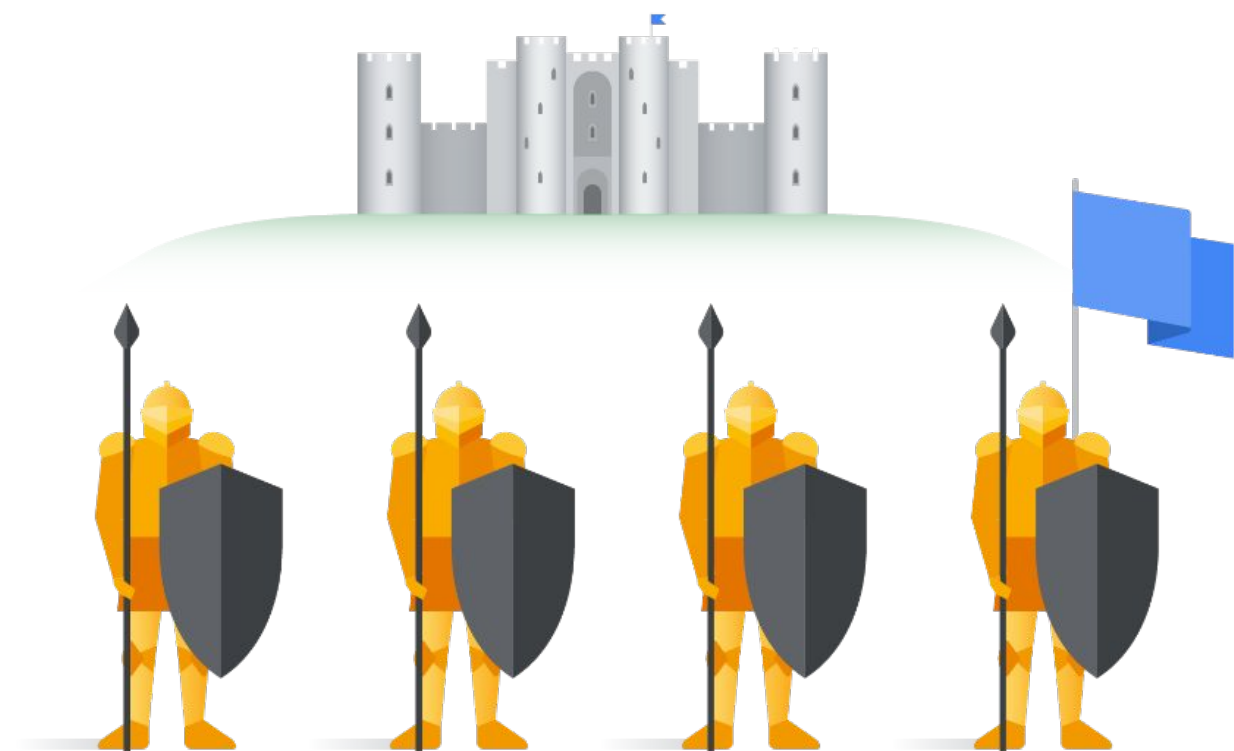


Cloud Logging

A service to collect and analyze security logs from your entire Google Cloud environment.



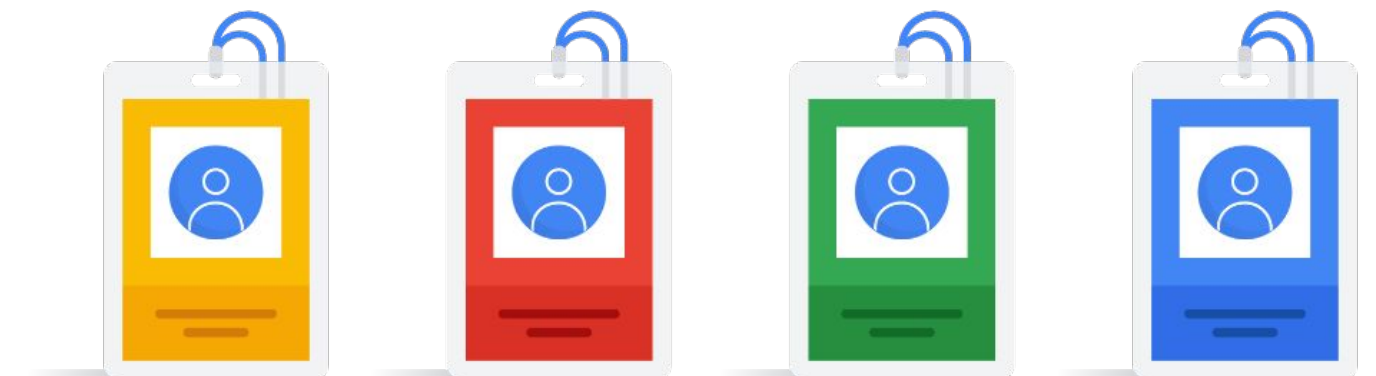
Incident response



It's like having a team of knights ready to defend your castle at a moment's notice.



Google Cloud



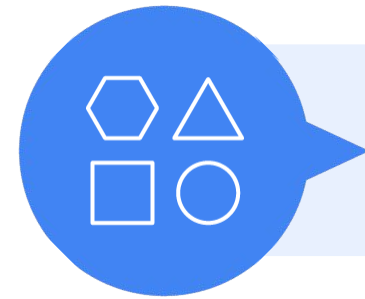
Google Cloud has expert incident responders across various domains.

It's important to educate employees on best practices

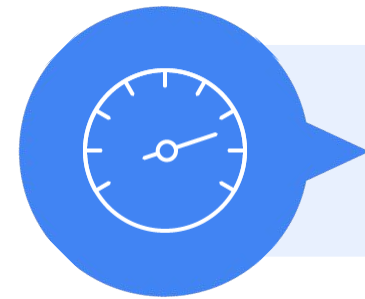


Security awareness training helps prevent incidents by raising awareness and empowering employees to protect themselves and the organization.

SecOps benefits



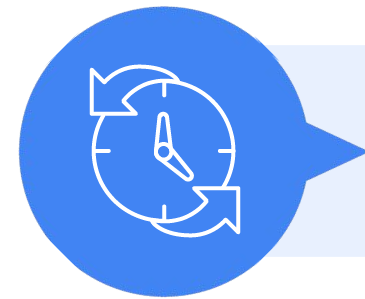
Reduced risk of data breaches



Increased uptime



Improved compliance



Enhanced employee productivity

Module 5

Trust and Security with Google Cloud



Lessons

- 01 Trust and security in the cloud
- 02 Google's trusted infrastructure
- 03 Google Cloud's trust principles and compliance

The Google Cloud trust principles

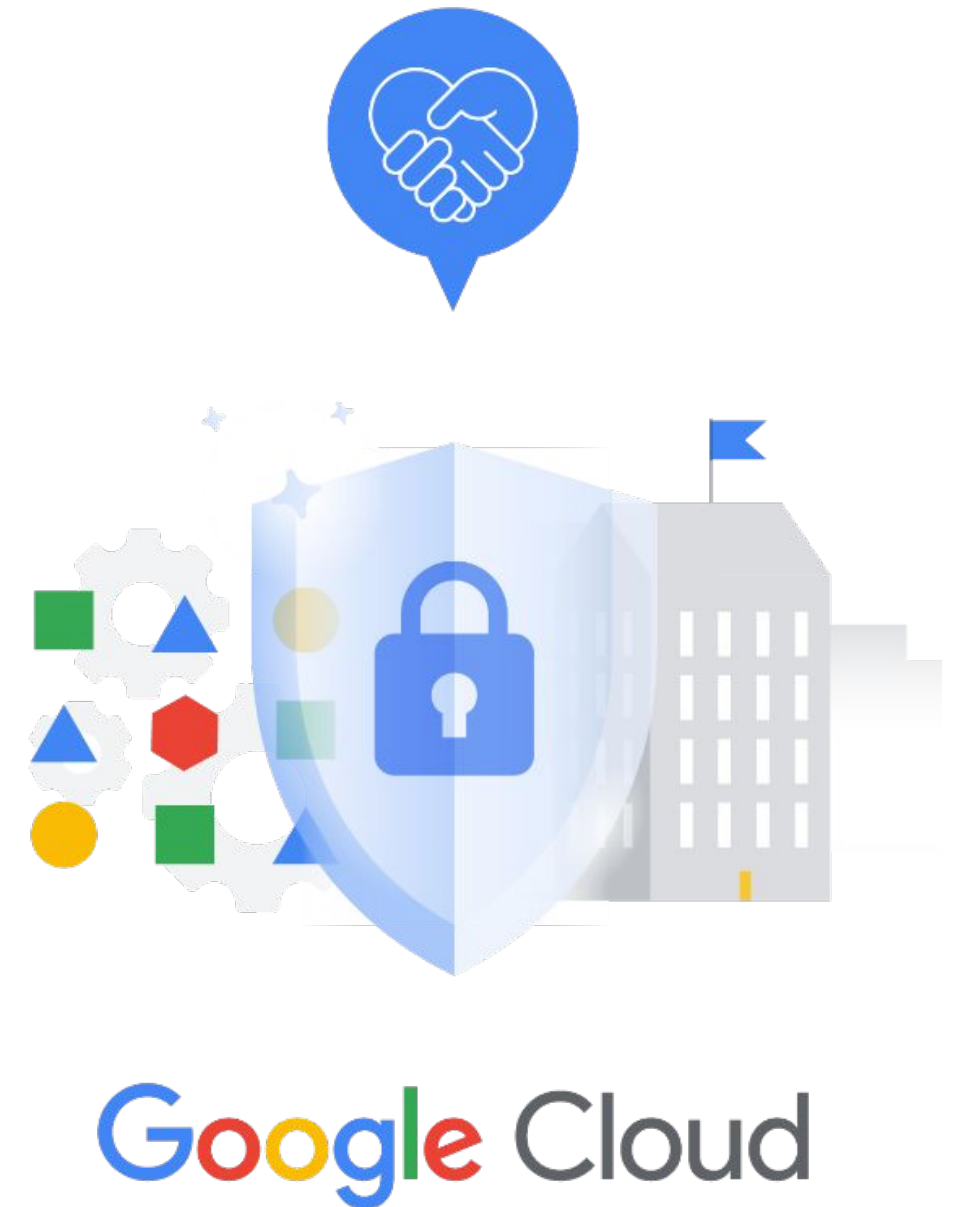


Google Cloud's trust principles are designed to empower you and ensure that the security and control of your business data is not compromised.

Google Cloud

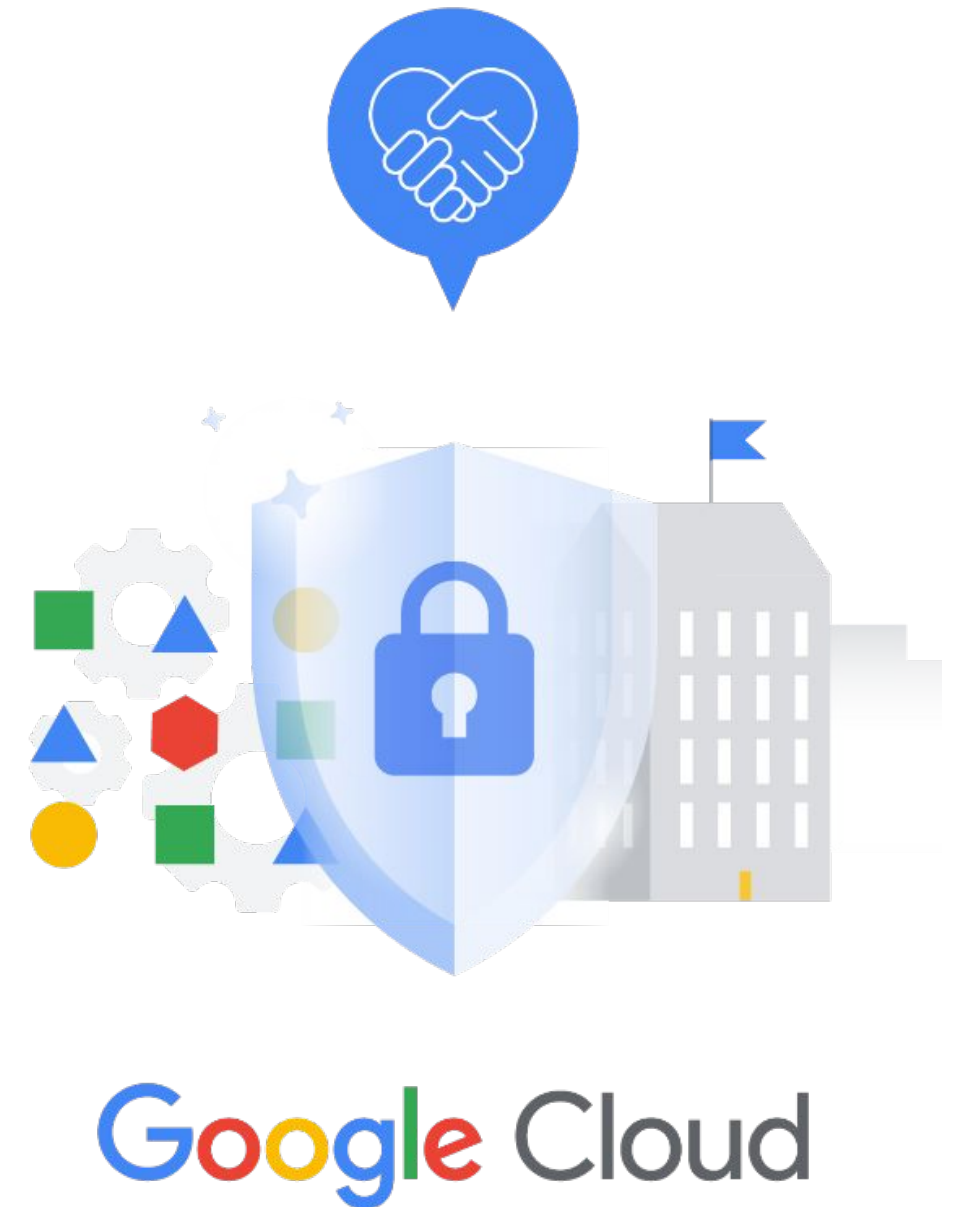
The Google Cloud trust principles

- 1 You own your data, not Google.
- 2 Google does not sell customer data to third parties.
- 3 Google Cloud does not use customer data for advertising.
- 4 All customer data is encrypted by default.
- 5 We guard against insider access to your data.
- 6 We never give any government entity "backdoor" access.
- 7 Our privacy practices are audited against international standards.



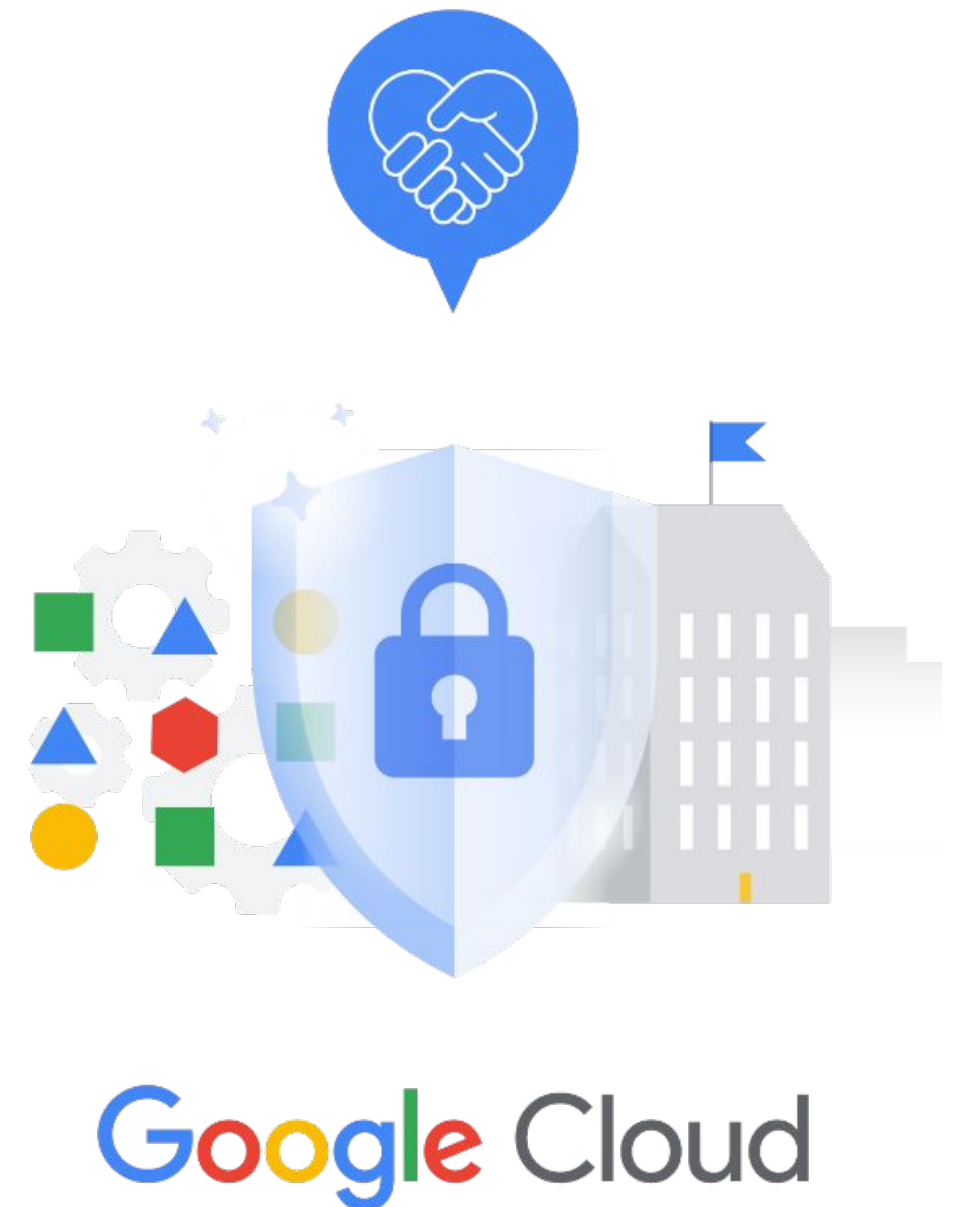
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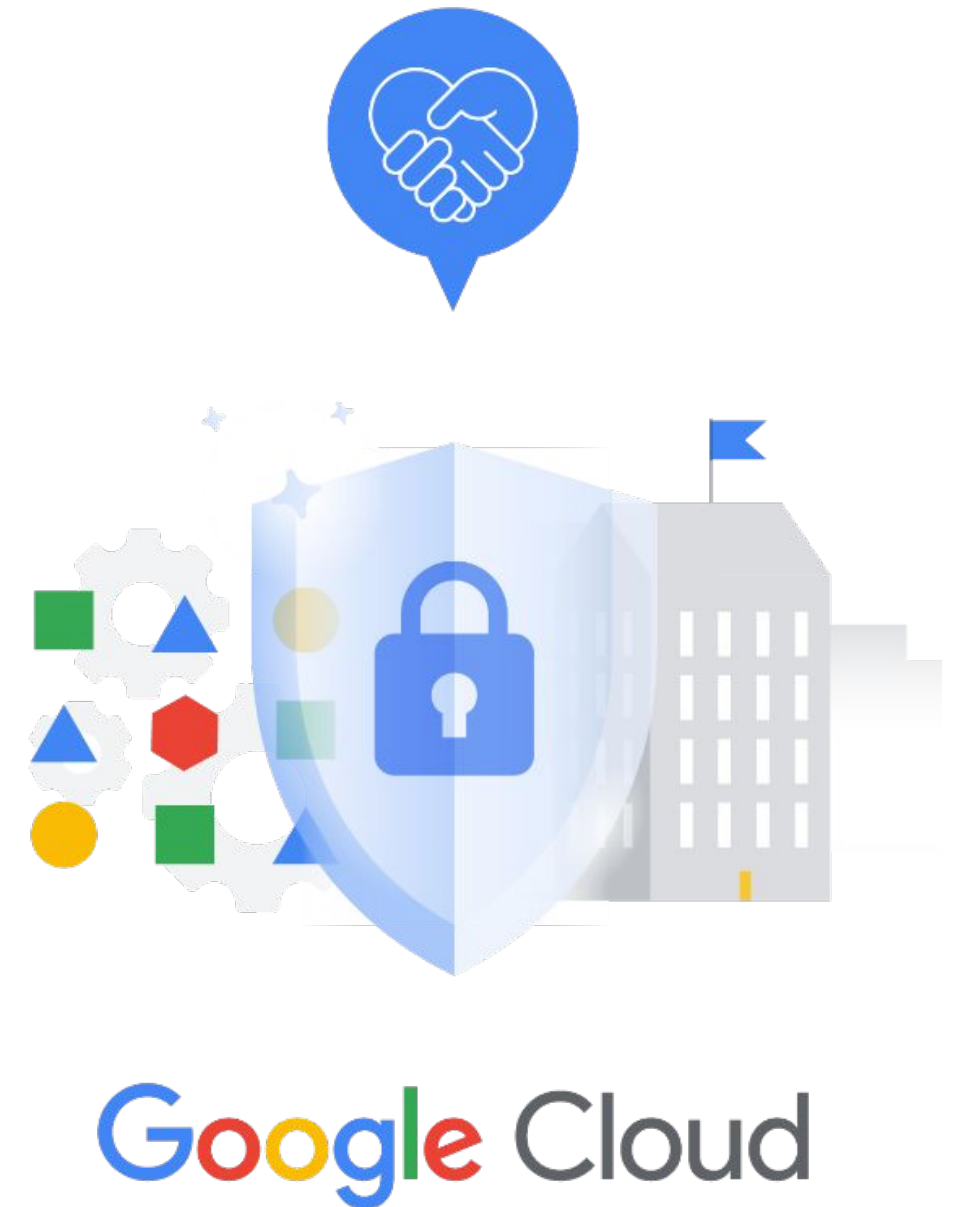
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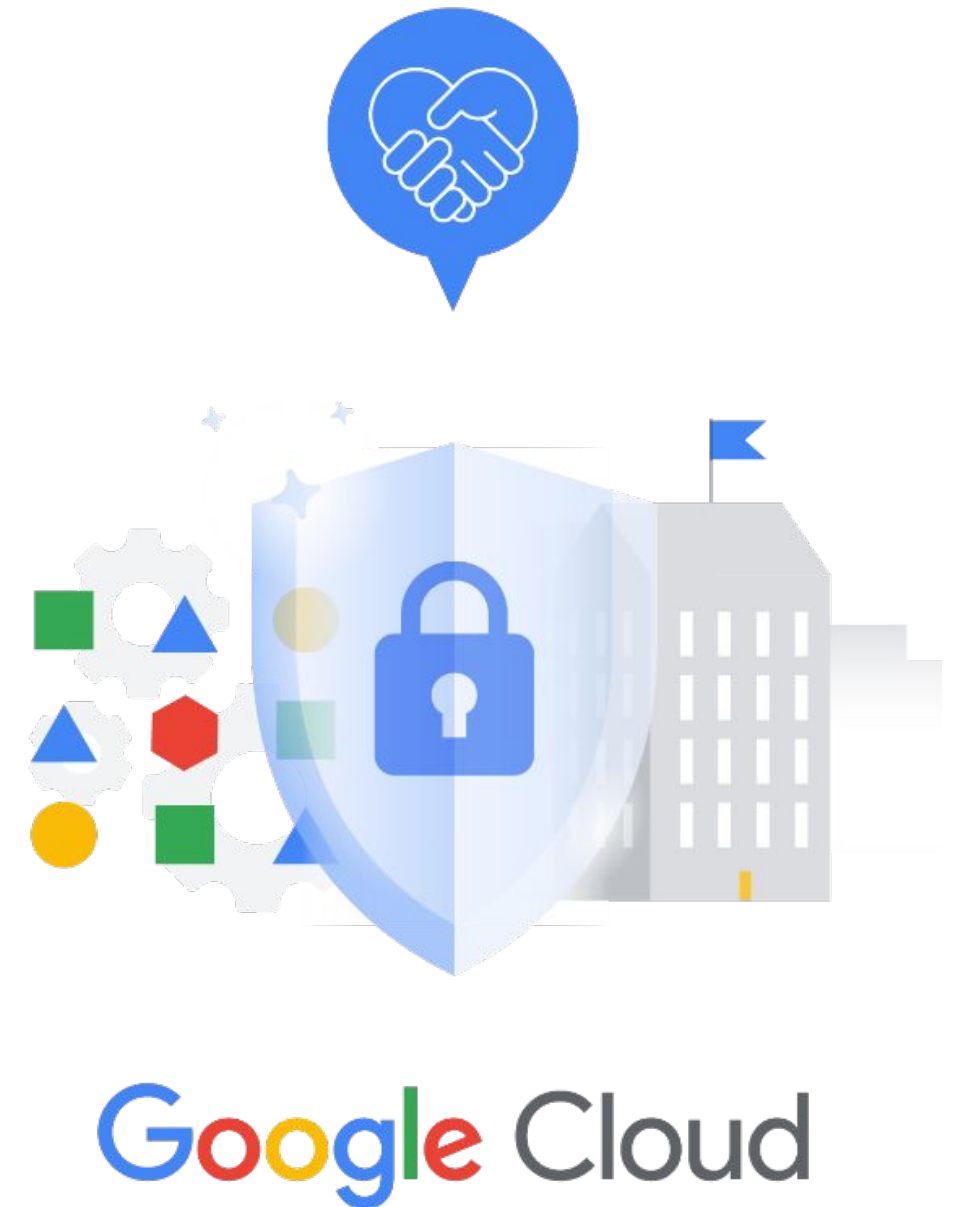
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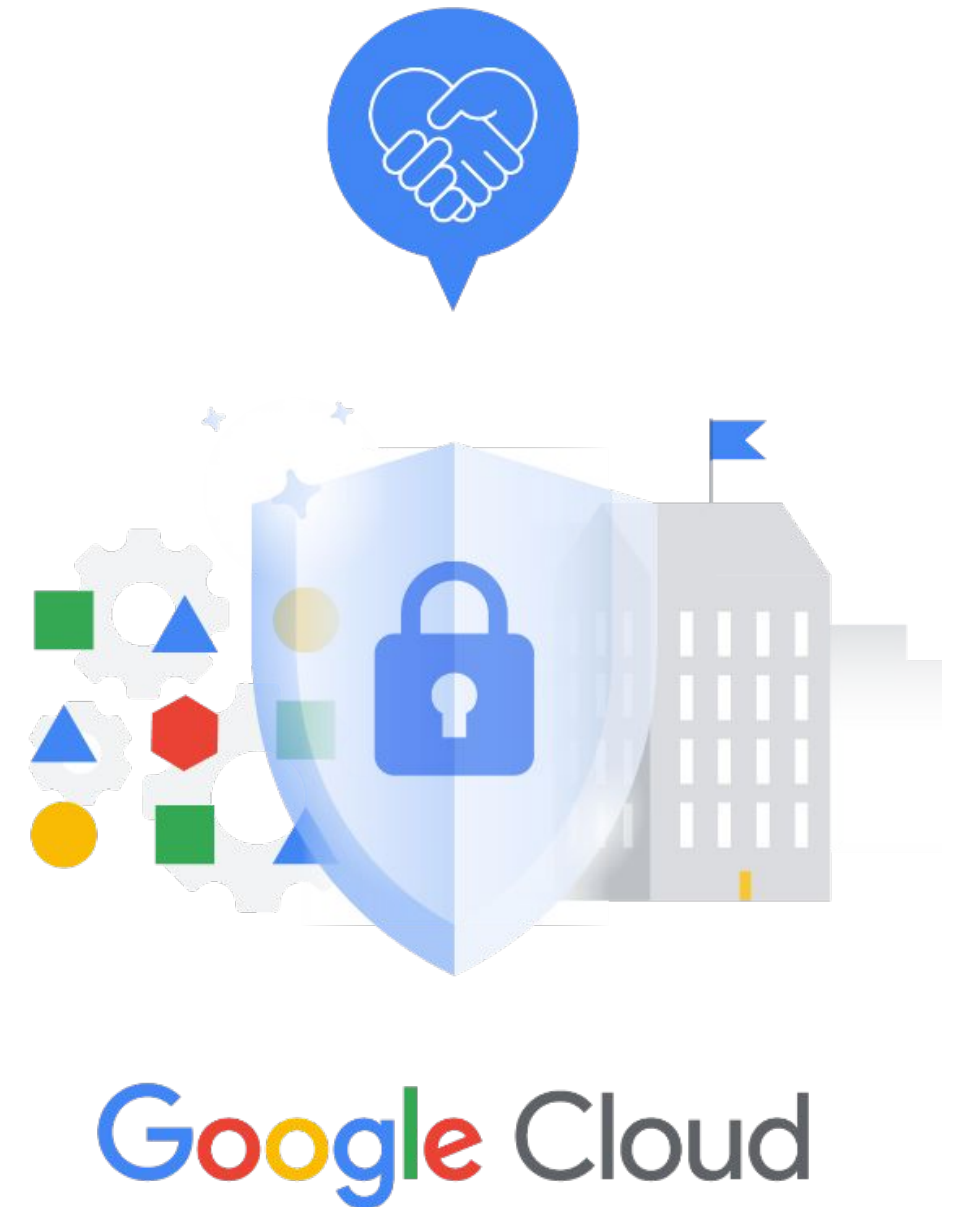
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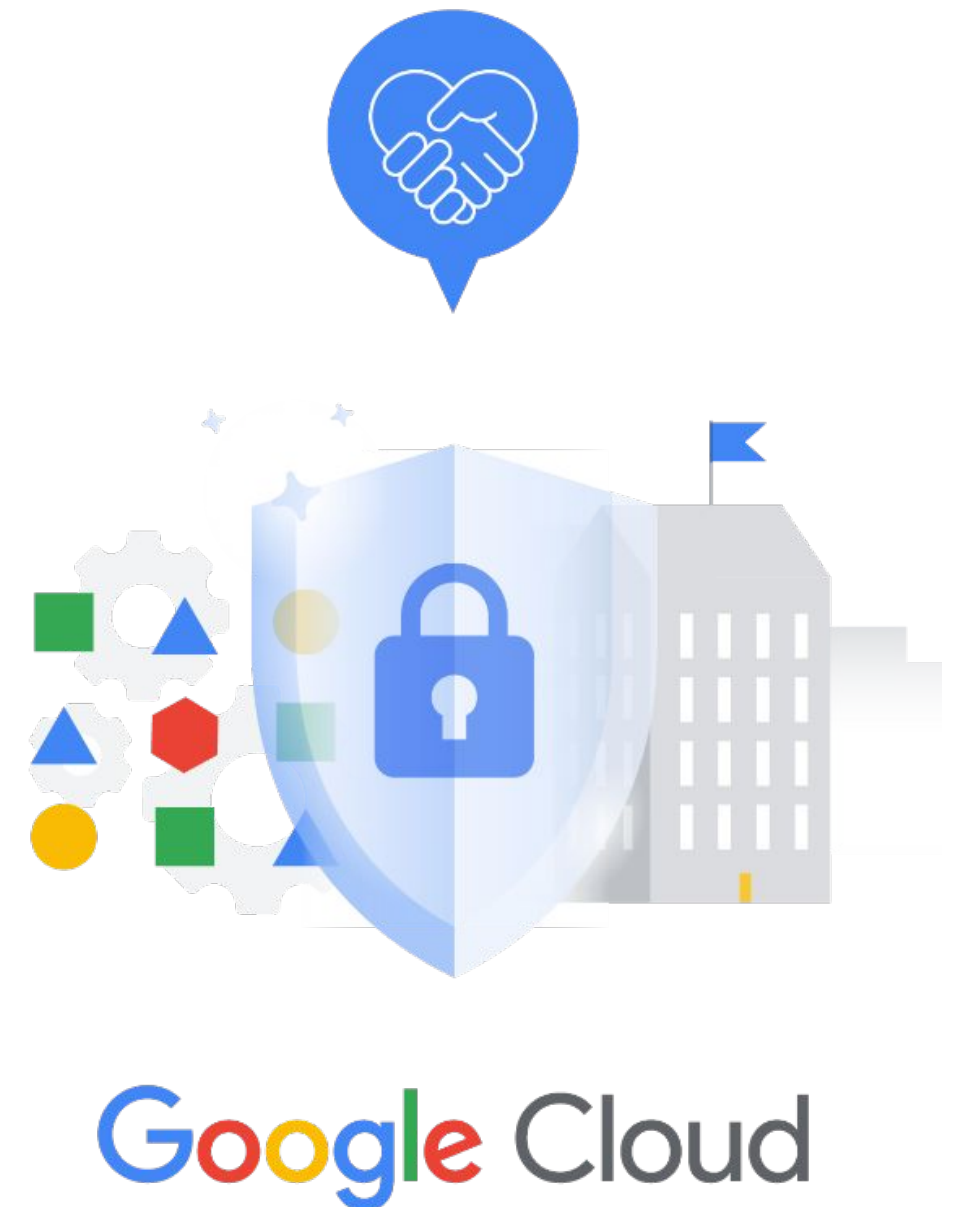
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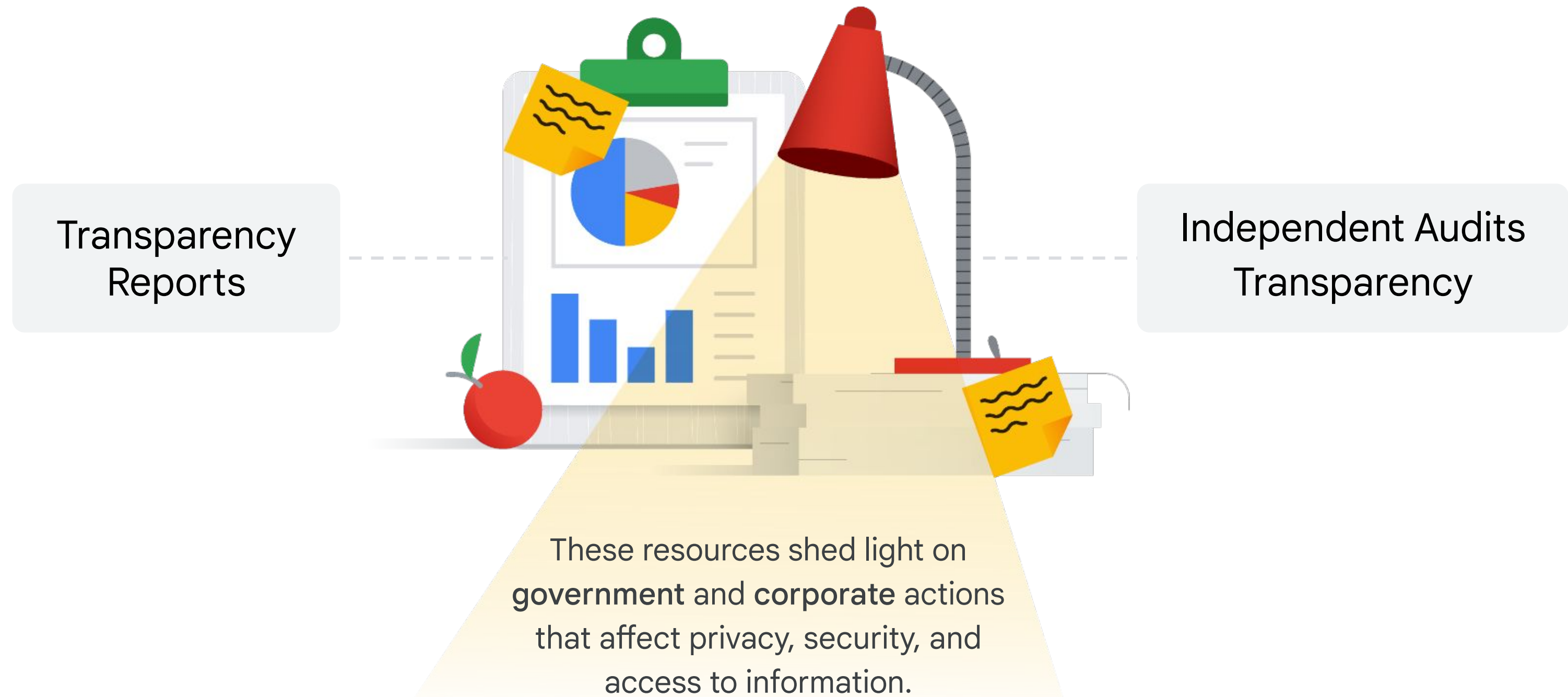


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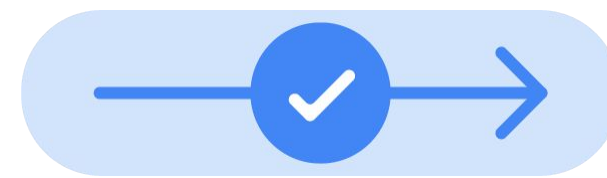
Reports to stay informed and maintain trust in Google Cloud services



Google Cloud undergoes independent, third-party audits and certifications

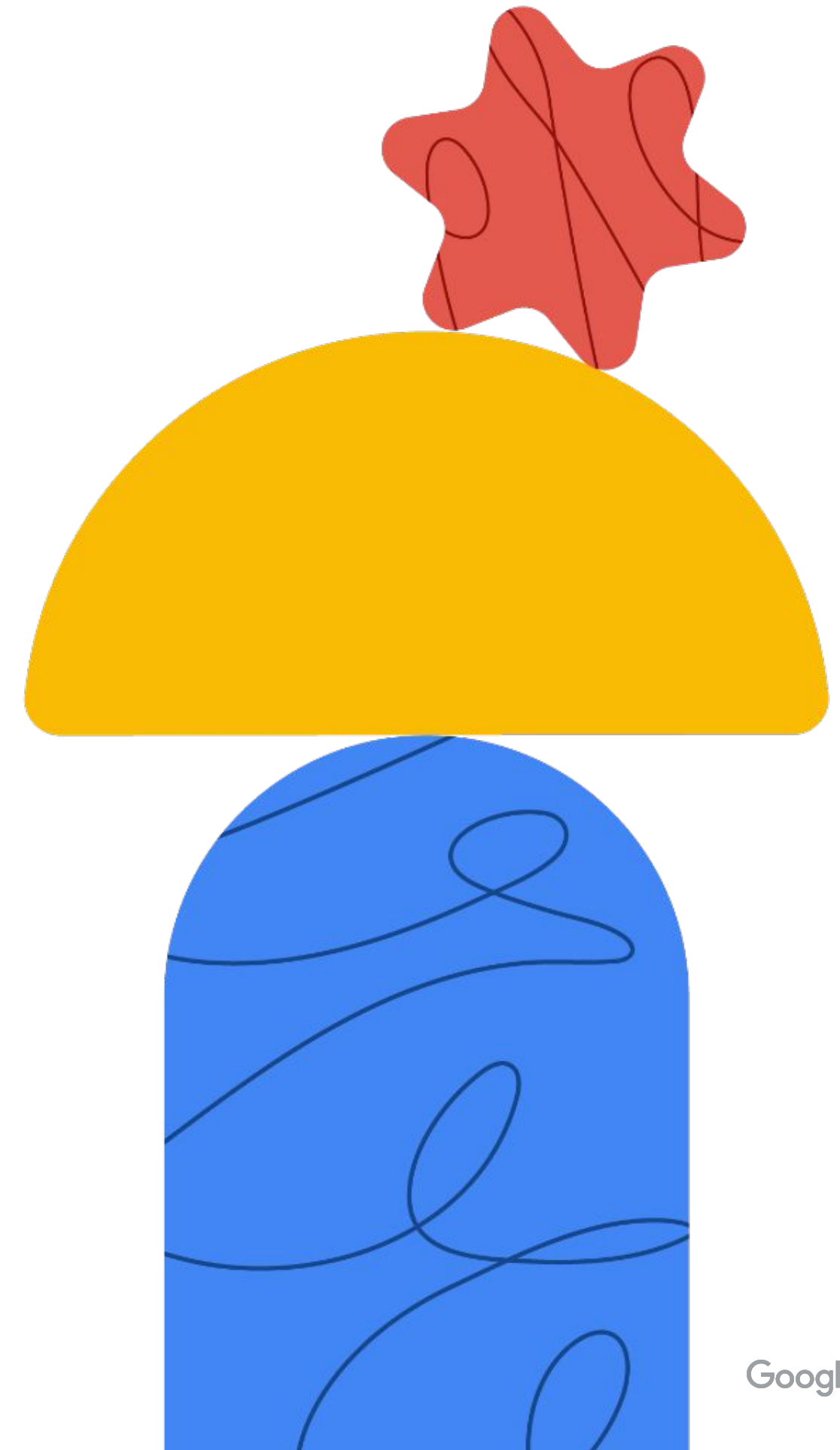


This verification process ensures that our data protection practices align with our commitments and industry standards.



Participation in initiatives like the EU Cloud Code of Conduct reinforces our dedication to accountability, compliance support, and robust data protection principles.

Data residency and data sovereignty



Data sovereignty

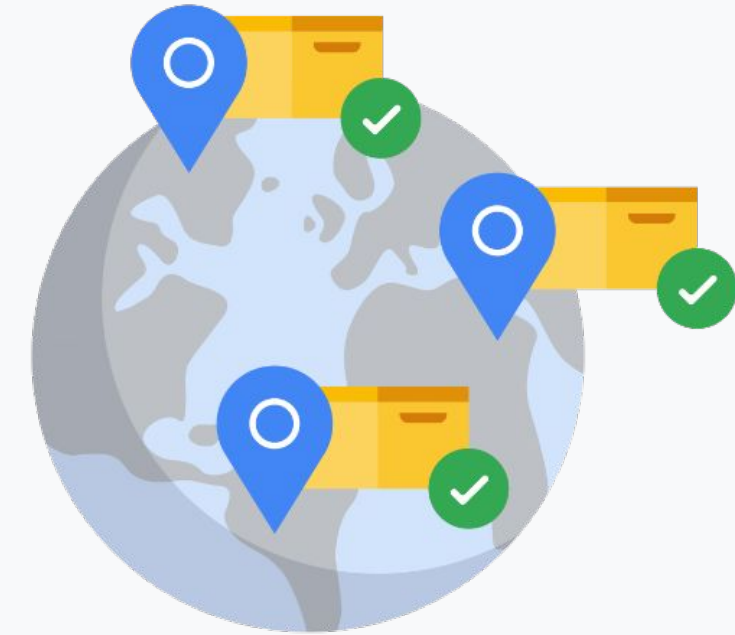
The legal concept that data is subject to the laws and regulations of the country where it resides.



The **General Data Protection Regulation** in the European Union requires companies to comply with data protection laws when processing or storing the personal data of EU citizens, regardless of their location.

Data residency

The physical location where data
is stored or processed.



Some countries mandate that the personal data of its citizens must be stored on servers within the country.

This ensures that data remains within the jurisdiction of local laws.

Google Cloud addresses data residency requirements by letting you choose where your data resides

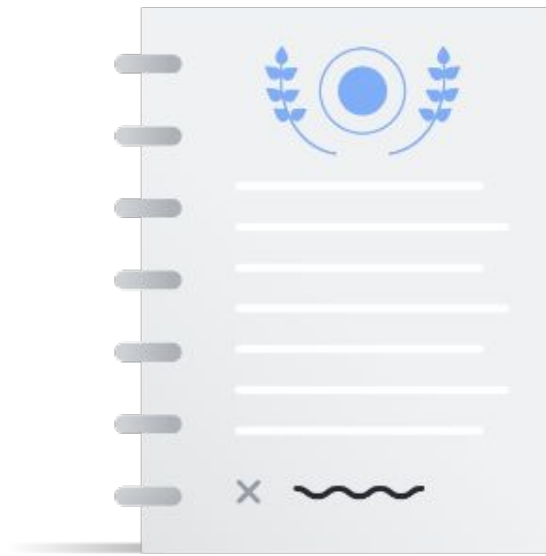
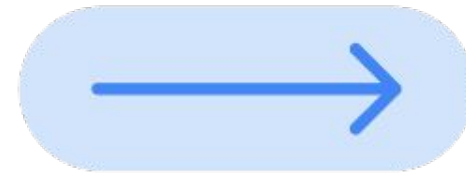
Within the European Union, you can select regions located in various countries like the UK, Belgium, Germany, Finland, Switzerland, and the Netherlands.



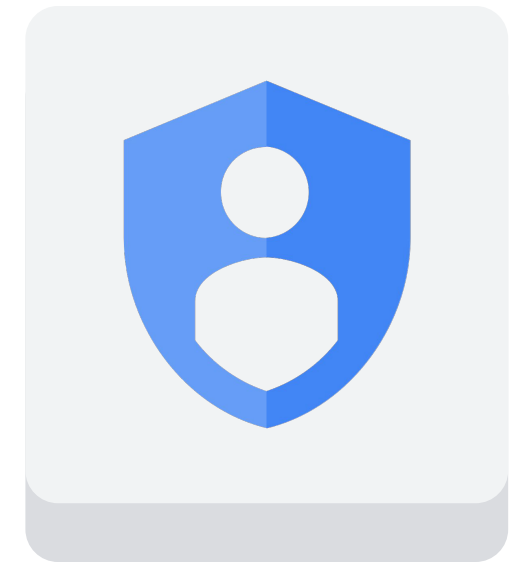
Organization policy constraints and IAM can help prevent accidental data storage in the wrong region



Google Cloud



Organization policy constraints



Identity and Access Management (IAM)

VPC Service Controls and Google Cloud Armor



VPC Service Controls

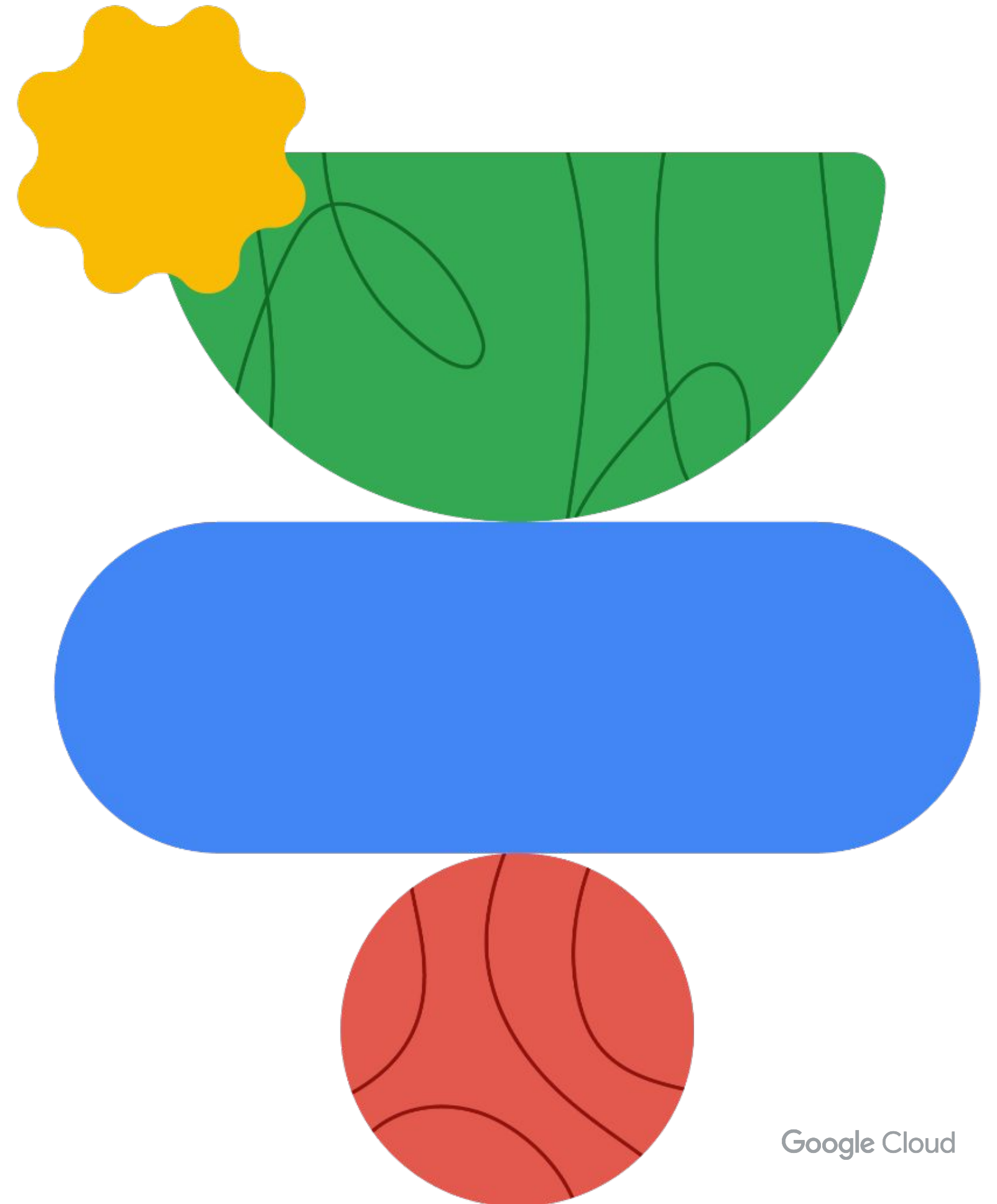
Restrict network access to data based on defined perimeters and limit user access through IP address filtering.



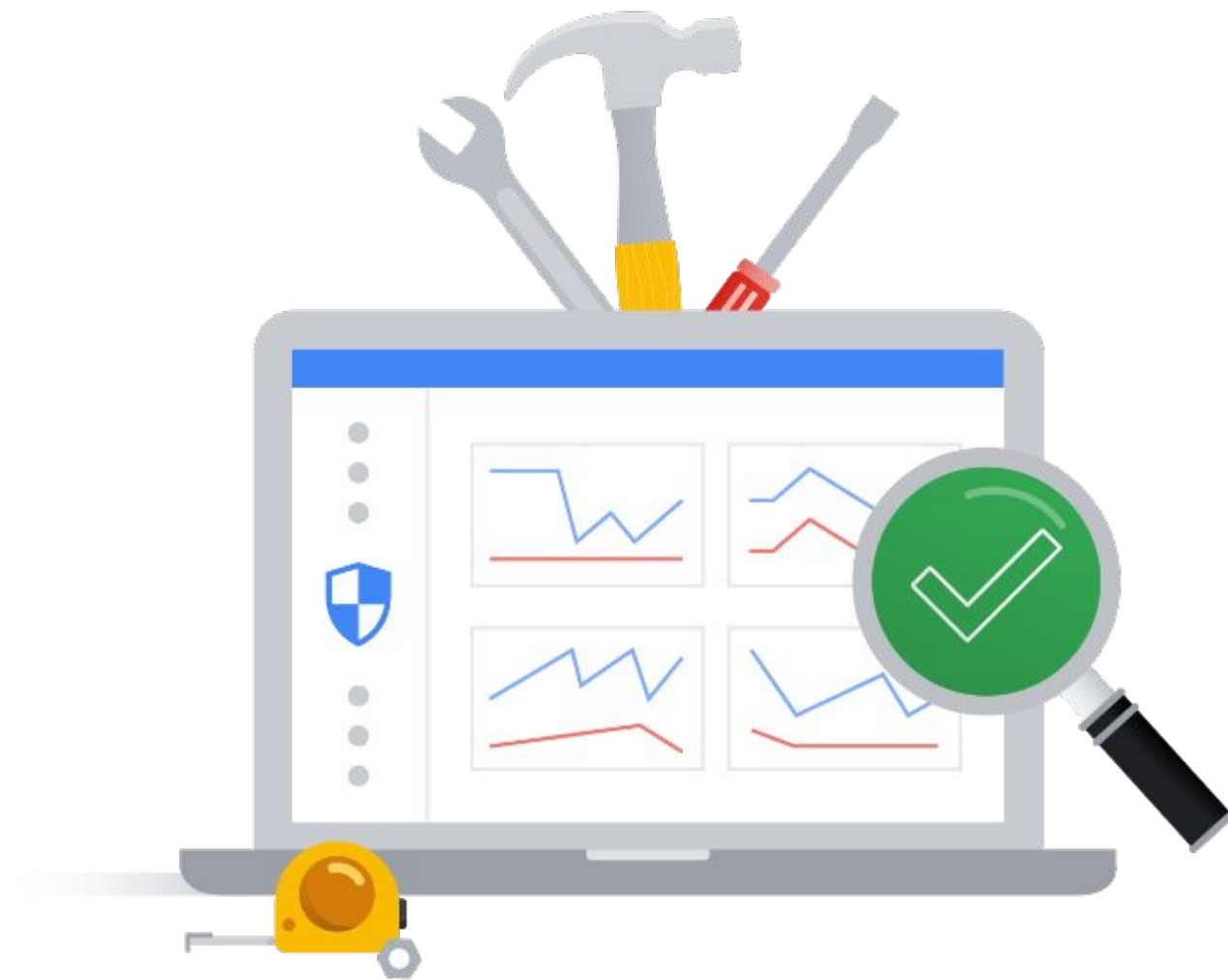
Cloud Armor

Restrict traffic locations for your external load balancer by adding an extra layer of protection.

Industry and regional compliance



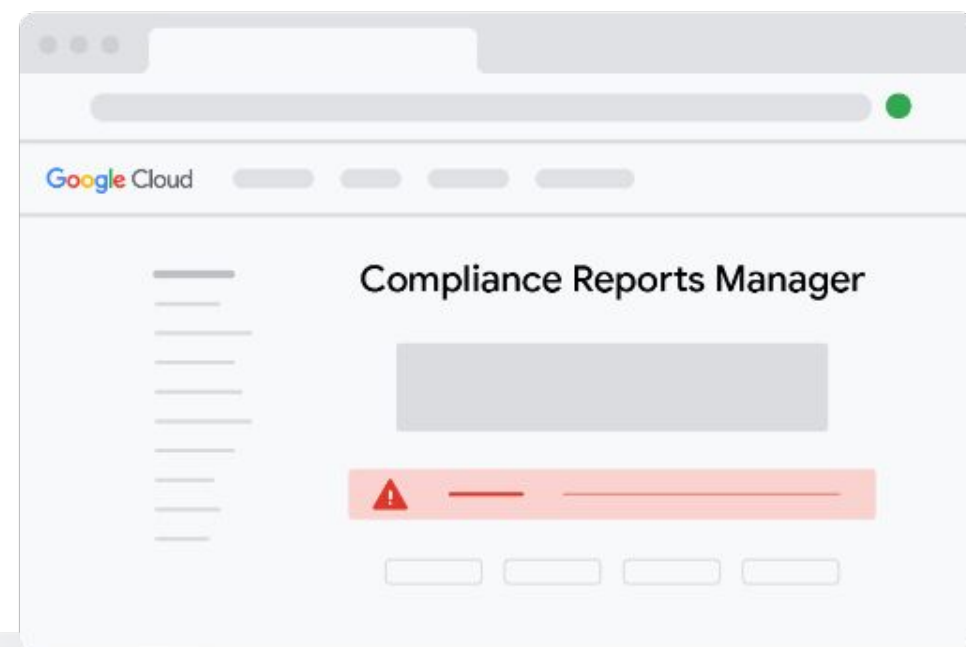
Resources and tools to help achieve compliance



Compliance resource center

- Provides detailed information on the certifications and compliance standards we satisfy.
- Offers valuable documentation on regional and sector-specific regulations.
- Equips you with the necessary insights and documentation to align your compliance efforts with HIPAA requirements.
- Provides guidance on meeting regulations in the financial sector like PCI DSS.

Resources and tools to help achieve compliance



Compliance Reports Manager



Offers easy, on-demand access to critical compliance resources at no extra cost.



Lets you discover Google's latest ISO/IEC certificates, SOC reports, and self-assessments.

Compliance resource center

cloud.google.com/security/compliance

Compliance Reports Manager

[cloud.google.com/security/compliance/
compliance-reports-manager](https://cloud.google.com/security/compliance/compliance-reports-manager)

Quiz

Question

Which is the responsibility of the cloud provider in a cloud security model?

- A. Setting access policies for the customer's data
- B. Maintaining the customer's infrastructure
- C. Configuring the customer's applications
- D. Managing the customer's user access

Quiz

Answer

Which is the responsibility of the cloud provider in a cloud security model?

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Quiz

Question

Which cybersecurity threat occurs when errors arise during the setup of resources, inadvertently exposing sensitive data and systems to unauthorized access?

- A. Phishing
- B. Virus
- C. Malware
- D. Configuration mishaps

Quiz

Answer

Which cybersecurity threat occurs when errors arise during the setup of resources, inadvertently exposing sensitive data and systems to unauthorized access?

- A. Phishing
- B. Virus
- C. Malware
- D. Configuration mishaps



Quiz

Question

What Google Cloud product provides robust protection from harmful distributed denial-of-service (DDoS) attacks?

- A. Google Cloud Armor
- B. Cloud Load Balancing
- C. Cloud IAM
- D. Cloud Monitoring

Quiz

Answer

What Google Cloud product provides robust protection from harmful distributed denial-of-service (DDoS) attacks?

- A. Google Cloud Armor
- B. Cloud Load Balancing
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Google Cloud